MDPI

Review

Training and Deployment Resources of Robot Vision–Language–Action Models: A Review of 215 Models with a Validated Feasibility Estimator

Firstname Lastname [1,*]

[1] Affiliation; yc424k@gmail.com

Featured Application

The open dataset and estimator let robot developers check, before downloading or training anything, whether a published VLA model fits the memory and control-rate budget of their onboard computer and how much GPU memory fine-tuning it will need.

Abstract

Vision–language–action (VLA) models have become a common way to build general robot policies, but the resources they need—training compute, inference latency and memory—are reported unevenly, which makes it hard to match a model to a robot computer or a training budget. We review 215 robot policy models released between October 2022 and October 2026 for manipulation, humanoid whole-body control, mobile manipulation and navigation. Each model was coded from primary sources into a 28-field schema with explicit labels for undisclosed and unverified values, and re-checked in a verification pass. Action heads shifted from discrete tokens to flow matching, and Qwen replaced Llama as the most common language backbone. Only 38 models (18%) report enough to compute training GPU-hours and 28 (13%) report inference memory; industry-only models disclose their training hardware less often than other teams (19% vs. 63%). We propose a roofline-based estimator of inference latency, memory and fine-tuning cost that models token decoding and iterative denoising. Calibrated on 37 published latency measurements from 11 devices, it reaches a median error of 1.37× (1.49× in leave-one-out validation). We use it to map which models can run at 10 Hz on eleven devices, from Jetson modules to an H100, and close with a ten-item reporting checklist.

Keywords: vision–language–action models; robot foundation models; embodied AI; training compute; edge deployment; inference latency; roofline model; reporting standards; humanoid robots; mobile manipulation

Received:
Revised:
Accepted:
Published:

Citation: Lastname, F. Training and Deployment Resources of Robot Vision–Language–Action Models: A Review of 215 Models with a Validated Feasibility Estimator. *Appl. Sci.* **2026**, *1*, 0. https://doi.org/

Copyright: © 2026 by the author. Submitted to *Appl. Sci.* for possible open access publication under the terms and conditions of the Creative Commons Attri- bution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Robot learning has converged on a recipe borrowed from language and vision: pretrain a large network on heterogeneous data, then adapt it to a robot and a task. Vision–language–action (VLA) models apply this recipe to control. They attach an action-generating head to a vision–language model (VLM), so that one network maps camera images and a language instruction to robot actions [1–3]. Since RT-2 [1] showed in 2023 that web-scale VLM pretraining transfers to robot control, VLAs have spread from tabletop manipulation to humanoid whole-body control [4,5], mobile manipulation [6,7] and navigation [8,9]. Academic laboratories and companies now release several such models per month (Figure 2a).

For anyone who wants to use one of these models, the first questions are practical. Will the model fit in the memory of the robot's onboard computer? Will it produce actions fast enough for the task? Can it be fine-tuned on the GPUs at hand, and what did it cost to pretrain? The answers depend on a handful of quantities—parameter counts, accelerator-hours, inference latency on a named device, peak memory—that are scattered across papers, model cards, repositories and blog posts, are reported in inconsistent units, or are not reported at all. Existing surveys organise the field by architecture and training paradigm [10–14], by efficiency technique [15,16], by the barriers to deployment at the edge [17], or by data and benchmarks [18]. They explain how VLAs work and how they might be made cheaper. They do not tell a reader how much a specific model costs to train or whether it will run on a specific device, and they do not measure how often the required numbers are missing.

This review takes a resource-centred view of the field. We coded 215 robot policy models released between October 2022 and October 2026 against a fixed schema drawn from primary sources, labelled every missing or uncertain value explicitly, and re-checked each record in a verification pass. On this corpus we make four contributions:

1. An open, evidence-labelled dataset of 215 models (135 VLAs, 58 visuomotor policies without a pretrained VLM and 22 related systems) covering manipulation, humanoid whole-body control, mobile manipulation and navigation, with 145 per-stage training-hardware records, 38 published latency measurements and a control-interface taxonomy for 98 mobile robots. The dataset is also published as an interactive gallery.
2. A quantitative synthesis of architecture and resource trends: the shift of action heads from discrete tokens to flow matching, the shift of language backbones from Llama to Qwen, model size, training compute and data, control rates, and where reinforcement learning (RL) enters the pipeline.
3. A disclosure audit of twelve resource items, which shows which numbers are reported, which are not, and how reporting differs between industry and academic teams.
4. A roofline-based estimator of inference latency, memory and fine-tuning cost, calibrated on the published measurements and validated by leave-one-out prediction, which we use to map how many models can run at 10 Hz on eleven devices from Jetson modules to an H100.

We end with a ten-item checklist for reporting the resources of robot foundation models.

Section 2 positions this review against prior surveys. Section 3 describes how the corpus was built and coded. Sections 4–6 present the corpus, architecture trends and training resources. Section 7 reports the disclosure audit. Section 8 presents the estimator and the feasibility maps. Sections 9 and 10 cover reinforcement learning and mobile-base control. Section 11 discusses implications and limitations, and Section 12 concludes.

2. Related Surveys

Early surveys of foundation models in robotics [19–21] predate most of the models in our corpus. They name real-time execution and compute as open challenges but do not quantify them. Surveys dedicated to VLAs [10–14,22–25] organise the literature by component (vision encoder, language backbone, action decoder), by action representation [12] or by training paradigm, and several discuss efficiency in qualitative terms. Two recent surveys make efficiency their subject. Guan et al. [15] group techniques for reducing latency, memory and training cost into model architecture, perception features, action generation and training or inference strategies, and Yu et al. [16] organise efficient-VLA methods across the model, training and data pipeline. Grover et al. [17] analyse the constraints of running embodied foundation models on edge hardware and argue that autoregressive VLAs are limited mainly by memory bandwidth and diffusion-based controllers by compute, a view

that our estimator formalises and tests against measurements. Other surveys cover datasets and benchmarks [18], world–action models [26,27], foundation models for mobile service robots [28] and VLAs for aerial robots and bimanual manipulation [29].

Table 1 summarises the focus of these surveys as stated by their authors. Prior surveys describe methods; this review treats resources as data. Judging from their stated scope, none of the surveys above codes per-model training hardware, inference latency and memory under explicit evidence labels, measures how often these values are disclosed, or validates a predictive latency model against published measurements.

Table 1. Recent surveys related to VLA models and the focus stated in their abstracts.

Survey	Year	Stated focus
Firoozi et al. [19]	2023	Foundation models across the robot autonomy stack
Hu et al. [20]	2023	Foundation models for general-purpose robots; meta-analysis
Ma et al. [10]	2024	VLA taxonomy: components, low-level control policies, high-level task planners
Sapkota et al. [11]	2025	Concepts, progress, applications and challenges (over 80 VLA models)
Zhong et al. [12]	2025	VLAs viewed through action tokenization
Shao et al. [13]	2025	Large-VLM-based VLAs for robotic manipulation
Kawaharazuka et al. [14]	2025	Full-stack review, including hardware, towards real-world applications
Guan et al. [15]	2025	Efficiency techniques for embodied manipulation
Yu et al. [16]	2025	Efficient VLAs across the model, training and data pipeline
Grover et al. [17]	2026	Deployment constraints of embodied foundation models at the edge
Wang et al. [18]	2026	Datasets, benchmarks and data engines
Sa et al. [29]	2026	VLAs for aerial robots and bimanual manipulation (183 contributions)
This review	2026	Training and deployment resources of 215 models in four domains; disclosure audit; latency, memory and fine-tuning estimator validated on published measurements

3. Materials and Methods

3.1. Scope and Inclusion Criteria

We included models first described between January 2022 and 19 October 2026 that map visual observations, optionally together with language and proprioception, to robot actions or to commands consumed by a robot controller, in one of four domains: manipulation, humanoid whole-body control, mobile manipulation and navigation. Each model was assigned one of three types.

- **VLA** (135 models): policies built on a pretrained VLM or language model, or presented by their authors as VLAs.
- **VA** (58 models): visuomotor policies trained without a pretrained VLM, such as ACT [30] and Diffusion Policy [31]. They are included as reference points because many VLAs reuse their action heads and because several remain strong baselines.
- **Other** (22 models): systems that are not single end-to-end policies but shape VLA practice, such as world–action models that predict future video and actions [32,33], embodied-reasoning VLMs that direct a VLA [34], inference-time techniques such as real-time chunking [35], and fleet data-collection systems [36].

The release date is the month of the first public description (paper, blog post or model card).

3.2. Identification of Models

Candidate models were collected from [TODO(author): e.g., the reference lists of prior surveys, arXiv cs.RO listings, and announcements by robotics companies and laboratories].

The corpus was assembled in three rounds with reference dates of 6 October 2026 (165 models, mostly manipulation and humanoid models), 14 October 2026 (46 mobile-manipulation and navigation models) and 19 October 2026 (four further releases). Values were coded as they stood on the reference date of each round.

3.3. Data Extraction

For each model, the 28 fields in Table 2 were filled from primary sources: the arXiv paper, the README of the official repository, the model card, or the developer's official blog post. Names of models, datasets and hardware were kept as written in the source. Three fields were also parsed as numbers for analysis: total parameters (billions), pretraining GPU/TPU-hours and data hours.

Table 2. The 28-field extraction schema.

Group	Fields
Identity	Model; group; type (VLA, VA, other); domain; release date; organisations; weights released
Architecture	Architecture; backbone and action representation; key features (two sentences); parameters (as reported); parameters (billions)
Training	Pretraining accelerator; pretraining duration; pretraining steps and batch; pretraining GPU/TPU-hours; basis and unit of the hours; fine-tuning accelerator and duration
Data and evaluation	Tasks and benchmarks; robots and embodiments; data type; data scale; data hours; main datasets
Inference	Inference device, rate, latency and memory as reported
Provenance	Notes; verification status; primary source URL

A second set of fields was coded from the same sources: the vision encoder; the language backbone and its family; the action head and its family (11 classes); action-chunk length; control rate; where inference runs (onboard, local PC, remote server or cloud); inference memory; whether and where RL is used (four statuses and six categories); the type of mobile base and the interface through which the policy commands it (eight classes); and links to official code and weight repositories. Training hardware was further normalised into per-stage records (stage, accelerator model, count, duration and the source's wording) for 109 models, giving 145 records.

3.4. Evidence Rules

Two labels distinguish missing values. "Not disclosed" means that no primary source reports the value; "unverified" means that the original could not be checked. Missing values were never estimated, with one exception: pretraining GPU-hours were computed as accelerator count times duration when both were reported as numbers. Values available only from secondary sources such as news articles or third-party papers were kept but marked (18 records contain at least one such value), values not checked against the original were marked separately (27 records), and conflicting values from different primary sources were recorded side by side. Every field keeps a link to its source.

3.5. Verification

Each record was then checked against its primary sources in a second, independent pass. Of the 215 records, 145 were confirmed unchanged, 58 were corrected, and 12 could not be fully verified (11 were not re-checked and one has no primary source). The verification status is stored with every record.

3.6. Derived Variables and Analysis

Models were grouped into four macro-domains from their coded domain (manipulation, humanoid, mobile manipulation including legged mobile manipulators, and navigation). Developers were grouped by affiliation into industry-only (74 models), academic-only (66) and mixed teams (73); public research institutes and non-profit laboratories (for example, Shanghai AI Laboratory, BAAI and the Allen Institute for AI) count as academic and corporate research laboratories (for example, Google DeepMind, NVIDIA, Microsoft Research and Toyota Research Institute) as industry; two models with no stated affiliation were left unclassified.

For the disclosure audit (Section 7) we scored twelve items per model as disclosed or not: total parameter count; backbone (V)LM, including an explicit statement that there is none; action-head type; training accelerator model for at least one stage; training duration for at least one stage; pretraining GPU/TPU-hours, reported or computable; amount of training data; control rate; inference device; inference memory; released weights (full or partial); and an official code repository. Differences between industry-only models and all other models were tested with Fisher's exact test and Holm's correction across the twelve items. We report medians and interquartile ranges, and rank correlations as Spearman's ρ. All numbers, tables and figures in this article are produced by scripts that read the released dataset directly.

3.7. Use of Generative AI

Generative AI was used in this work as follows: [TODO(author): describe any use in locating sources and extracting or verifying values]. A generative AI assistant (Claude, Anthropic) was also used to write the analysis and plotting scripts and to draft the text of this manuscript. All analyses were run on the released dataset, and the author reviewed and edited all text and takes full responsibility for the content.

4. The Corpus

Figure 1 shows the anatomy that the schema follows. A VLA encodes one or more camera views, passes the visual tokens and the instruction through a (V)LM backbone, and decodes actions with a head; the head either emits discrete tokens one at a time, runs several denoising passes of a smaller network, or predicts actions in one forward pass. Many systems, especially humanoids and legged robots, add a low-level controller that turns the policy's output into joint commands.

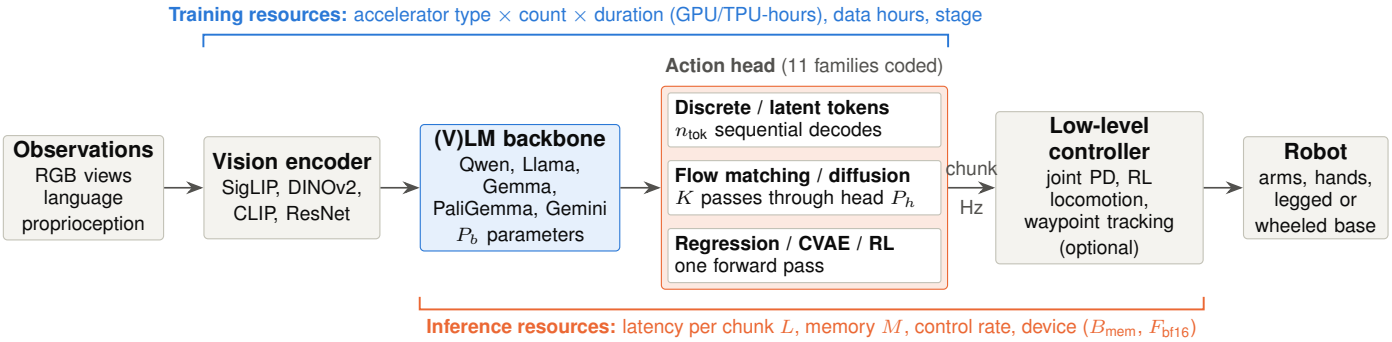


Figure 1. Anatomy of a VLA system and the resources coded in this review. Training resources cover the vision encoder, backbone and action head; inference resources span the backbone, the action head and any low-level controller. The symbols are those of the estimator in Section 8.

The corpus grows from 2 models in 2022 and 14 in 2023 to 60 in 2024 and 91 in 2025; 48 models appeared in 2026 up to the cut-off of 19 October (Figure 2a). Manipulation accounts

for 111 models, humanoid whole-body control for 41, mobile manipulation for 34 and navigation for 29. Industry-only teams released 74 models, academic-only teams 66 and mixed teams 73.

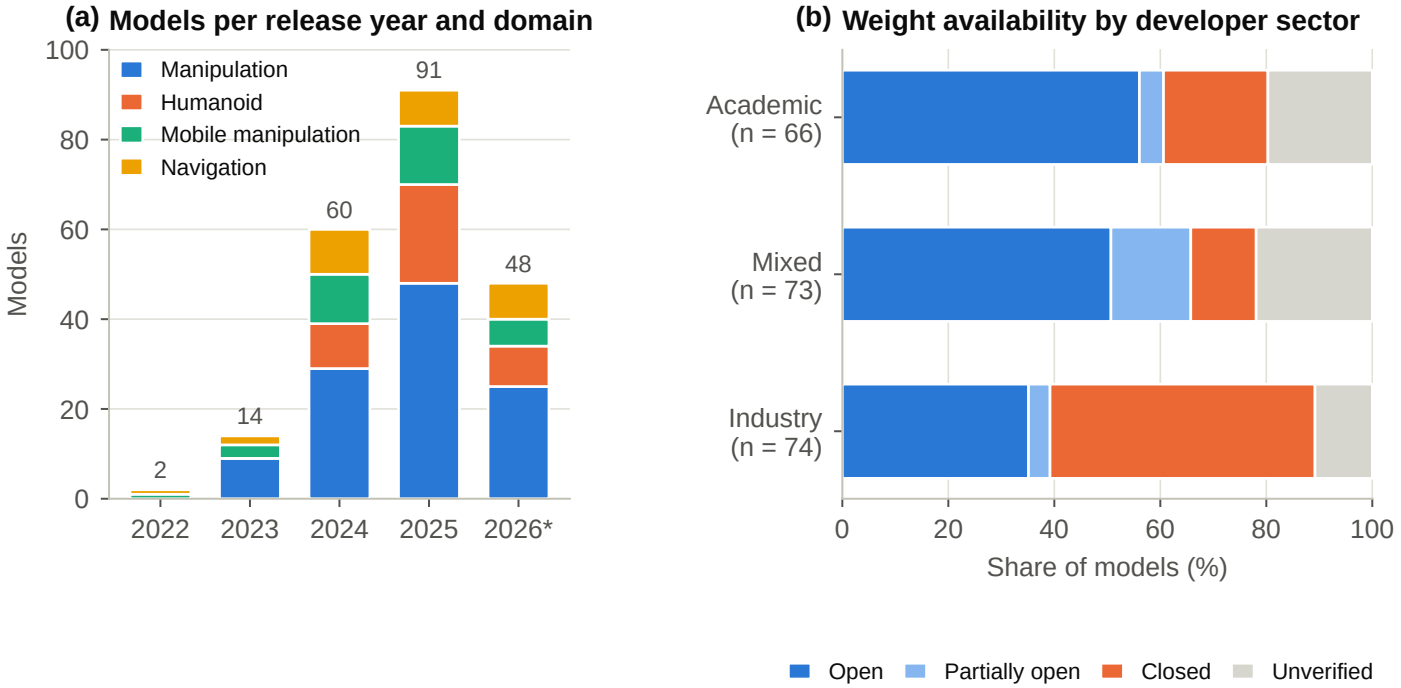


Figure 2. Composition of the corpus. (**a**) Models per release year and domain (*2026 up to 19 October). (**b**) Weight availability by developer sector.

Most models are documented by a paper: 182 records cite an arXiv paper (176 distinct preprints, since some papers describe two models), 3 cite a technical report, and 30 are documented only on the web, through blog posts, model cards, repositories or, in one case, news articles. Web-only documentation is concentrated in industry: 29 of the 74 industry-only models (39%) have no paper or report, against one of the 66 academic-only models. Weights are fully released for 100 models and partially for 17, withheld for 59, and could not be verified for 39. 89 models have weights on Hugging Face and 153 have an official code repository. Openness also follows the developer: 61% of academic-only and 66% of mixed-team models release weights at least partially, against 39% of industry-only models; half of the industry-only models (37 of 74) are closed (Figure 2b).

5. Architecture Trends

5.1. Action Heads

The action head is where VLA designs have changed most (Figure 3a). Early VLAs followed RT-1 and RT-2 in discretising each action dimension into bins and decoding them as language tokens [1,2,37]. Discrete or latent-action tokens still made up 30% of the models released in 2024 (18 of 60) but 21% in 2025 (19 of 91) and 8% in 2026 (4 of 48). Flow matching [38], introduced to VLAs by π_0 [3], went the other way: one model in 2024, 18 in 2025 (20%) and 17 in 2026 (35%), when it became the most common disclosed head. Diffusion heads [39], popularised for visuomotor control by Diffusion Policy [31], were common in 2024 (15 models) and rare in 2026 (one model). Across the whole corpus the head families are discrete tokens (41), flow matching (36), diffusion (33), regression (26), RL policies (18), hybrids of autoregressive and diffusion or flow decoding (9), video world models (7), latent actions (6), CVAE (4) and other (7); 28 models do not disclose the head.

The undisclosed share grew from 3% in 2024 to 33% in 2026, a first sign of the reporting gap analysed in Section 7.

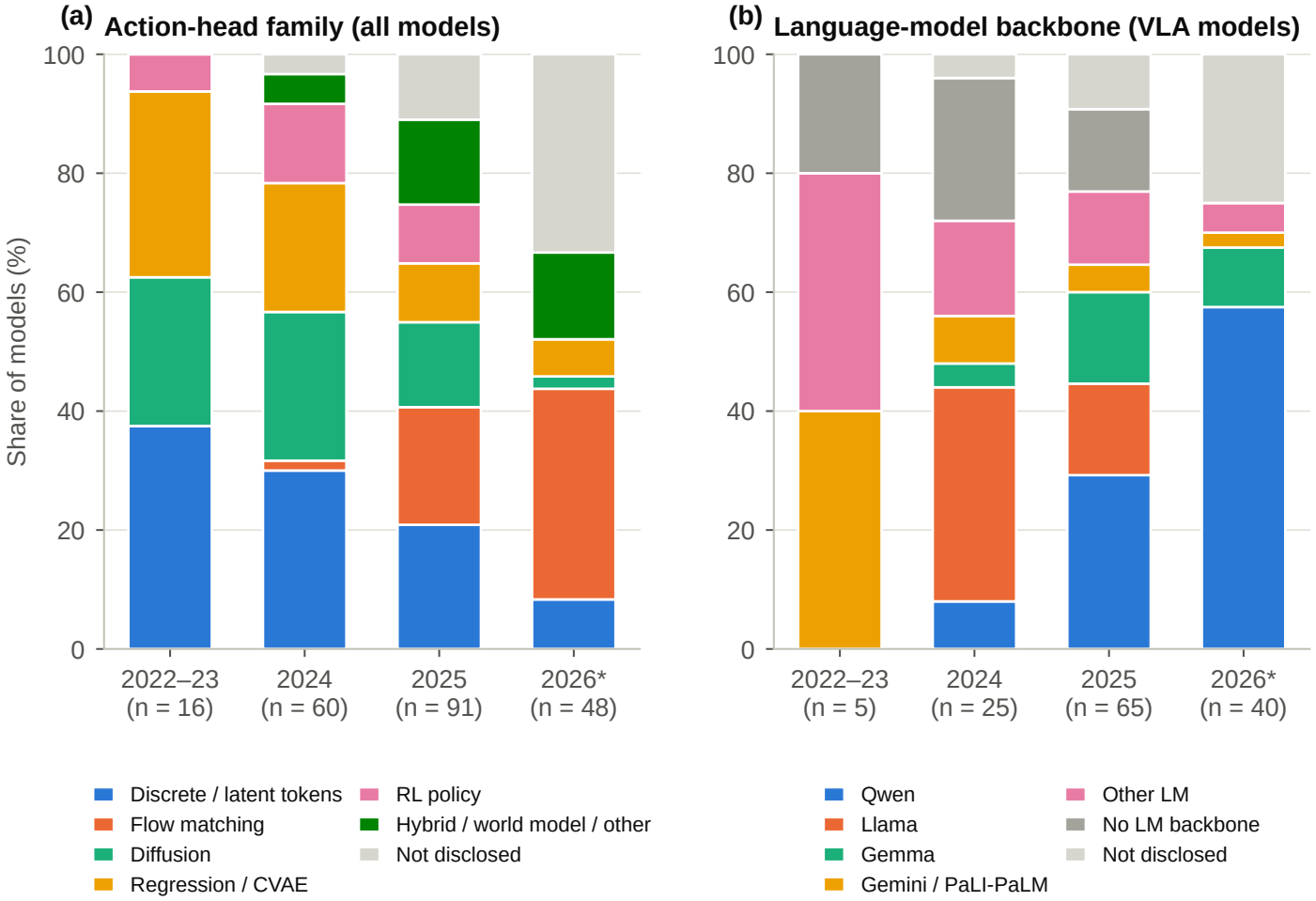


Figure 3. Architecture trends by release year (*2026 up to 19 October). (**a**) Action-head family across all models. (**b**) Language-model family of the backbone among VLA models.

5.2. Language Backbones

Among the 135 VLAs, the Qwen family is the most common backbone (44 models), followed by Llama (19), Gemma including PaliGemma [40] (15), Gemini (5) and PaLI/PaLM (3); 16 use other language models (SmolLM, InternLM, Phi, Mamba and others), 16 condition on language through a separate text encoder without a language-model backbone (for example, RDT-1B [41]), and 17 do not disclose the backbone. The mix changed quickly (Figure 3b). In 2024, Llama-based backbones such as the Prismatic VLM [42] behind OpenVLA [2] accounted for 9 of 25 VLAs and Qwen for 2. In 2025 Qwen led with 19 of 65, and in 2026 it accounts for 23 of 40 VLAs while no new VLA in the corpus uses Llama. One likely driver is the availability of strong small Qwen VLMs [43], but the corpus cannot test this.

Where the size of the language model is reported (81 models), the median is 4B. The most common size is 7B (26 models), 42 models use 4B or less and five use 32B or more. Most models with a reported language-model size (68 of 81, nearly all of them VLAs) use 2–8.5B parameters, which matters for the memory analysis in Section 8.

5.3. Model Size

Total parameter counts are disclosed for 113 models (median 3.3B; Figure 4). VLAs are much larger than visuomotor policies without a VLM (median 5B versus 0.093B). The yearly median rose from 0.0975B in 2022–2023 to 1.25B in 2024 and 4.1B in 2025, and stayed at

4.0B in 2026. The largest models are the 55B RT-2 variants [1,44,45] and the 72B DiVLA [46], but no model released since January 2025 discloses more than 14B parameters. The plateau around 4B is consistent with designs that must run on a robot or a single GPU at control rates; it may also reflect that the largest proprietary models do not disclose their size.

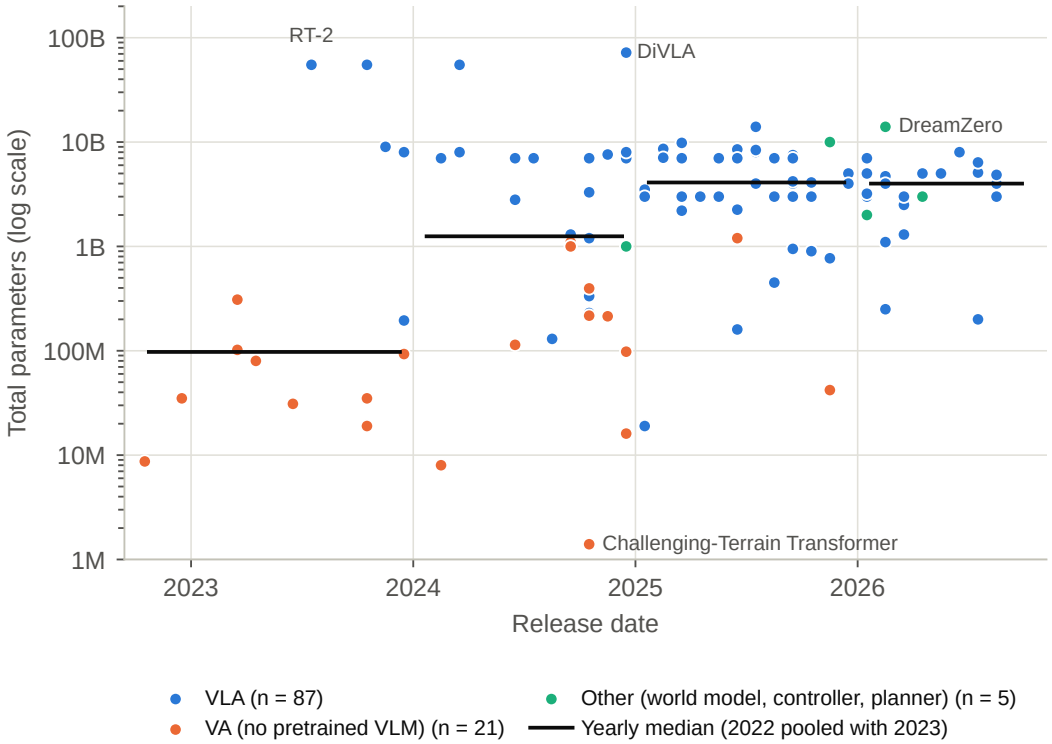


Figure 4. Total parameters against release date for the 113 models that disclose them. Black segments mark the yearly median.

5.4. Chunking, Control Rate and Hierarchy

Of the models that describe their action output, 92 predict chunks of several future actions and 20 predict one step at a time; 103 do not say. Chunking decouples the control rate from inference latency: a policy that needs 70 ms per inference can still command the robot at 50 Hz by executing a 50-step chunk while the next one is computed [3,30,35]. Eighty-eight models report a control or execution rate (median 20 Hz among the 83 with a numeric value; 58 at 10 Hz or more). The rate depends on the head: models with discrete-token heads report a median of 3 Hz (11 models), against 10 Hz for diffusion (20), 30 Hz for flow matching (11) and 50 Hz for RL policies (13).

Hierarchy is the other route to speed. Forty-nine models document a multi-stage pipeline; 28 include a separate low-level controller and 28 include a VLM or LLM stage. Typical are dual-system designs in which a slow VLM runs at a few hertz and a small, fast action module runs at tens to hundreds of hertz: GR00T N1 pairs a 10 Hz VLM with a 120 Hz flow-matching transformer [4], and Helix pairs a 7B VLM at 7–9 Hz with an 80M-parameter policy at 200 Hz [5].

6. Training Resources

6.1. Accelerators by Training Stage

The 145 training records from 109 models cover pretraining (76 records), post-training (2), fine-tuning (30), per-task training from scratch (16) and RL in simulation (21) (Figure 5b). Pretraining runs almost entirely on data-centre accelerators: Hopper-class GPUs (H100, H800 and H200; 26 records) and A100/A800 (23) dominate, followed by TPUs (8), V100 (4),

workstation GPUs (4) and GB200 (2); nine records do not name the accelerator. Where the count is reported (59 records), the median pretraining job used 32 accelerators; 21 used 64 or more, up to 1,024 H100 GPUs for GR00T N1 [4], and 23 used eight or fewer. The median reported pretraining duration is about four days (94 h, 36 records), ranging from 14 h for Octo-Base on a TPU v4 pod [47] to more than eight months for GEN-1.5 [48].

Adaptation is far cheaper. Fine-tuning and per-task training used a median of eight GPUs (45 records with a count; 39 used at most eight and 10 a single GPU) for a median of 24 h (30 records). Simulation RL, used mainly to train locomotion and whole-body controllers, typically runs on a single GPU (median of 20 records); 15 of the 21 records use workstation or consumer GPUs, in line with GPU-parallel simulators such as Isaac Gym [49].

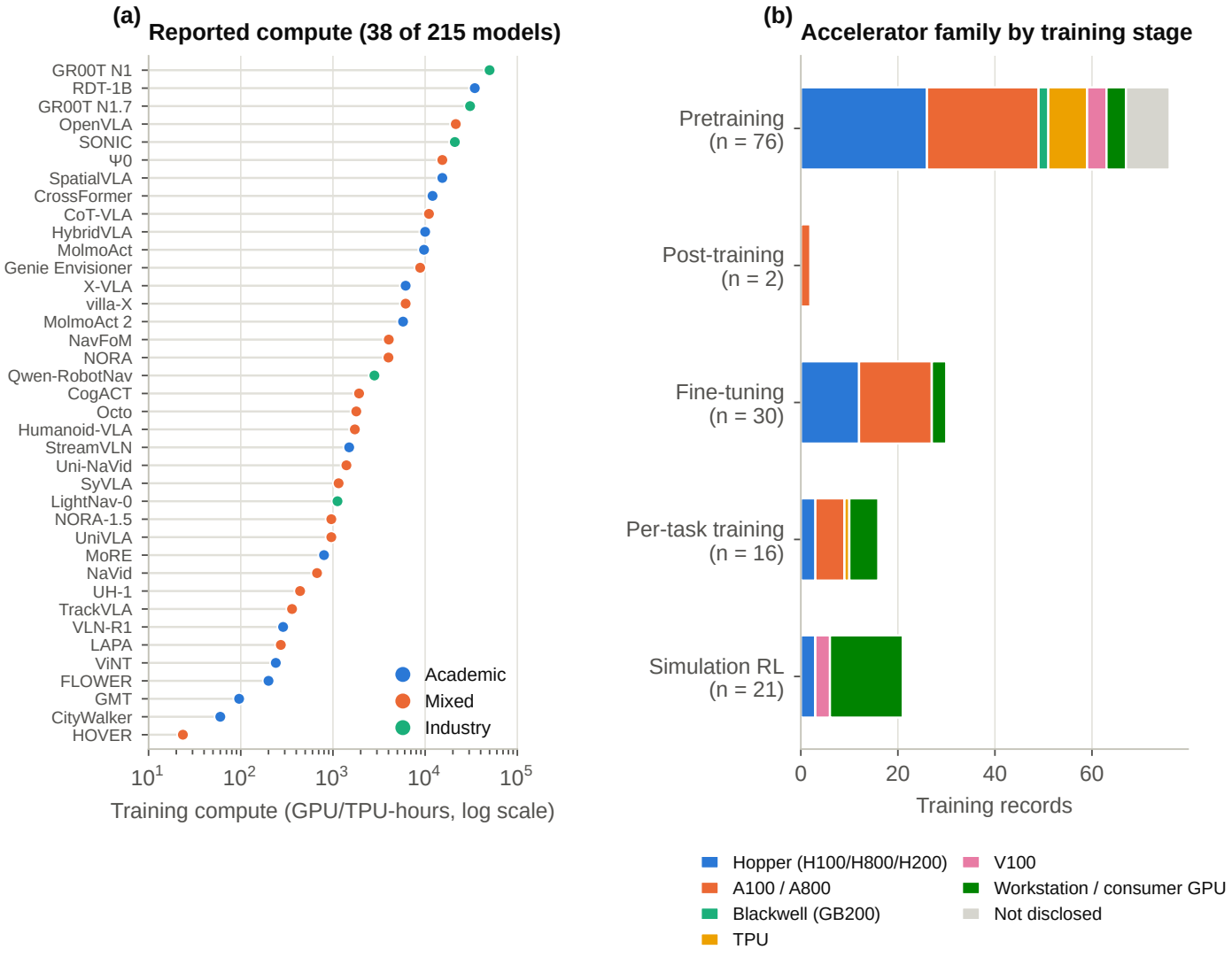


Figure 5. Training resources. (**a**) Pretraining GPU/TPU-hours for the 38 models where they are reported or computable, coloured by developer sector. (**b**) Accelerator family of each training record by stage.

6.2. Training Compute

Only 38 models report pretraining compute or both of its factors (Figure 5a). Their median is 1,856 GPU/TPU-hours (interquartile range 704–9,932), from 24 hours for HOVER [50] to roughly 50,000 H100-hours for GR00T N1 [4]. Ten models used 10,000 hours or more: besides GR00T N1, these include RDT-1B (about 34,560 H100-hours [41]), GR00T N1.7 (30,720 GB200-hours [51]) and OpenVLA (21,500 A100-hours [2]). Thirteen used fewer

than 1,000 hours. Compute is only weakly related to model size ($\rho = -0.18$, $n = 30$) and to release date ($\rho = 0.16$, $n = 38$), but it rises with the amount of training data among the 12 models that report both ($\rho = 0.64$). Data volume and training length, rather than parameter count, appear to set the cost, although these samples are small. These hours mix accelerator generations, from V100 to GB200 and TPU, and are not normalised to a common unit; they are a lower bound on the true cost because they exclude ablations, failed runs, data processing and simulation.

6.3. Training Data

Real-robot data appear in 145 models (mostly teleoperation), simulated or synthetic data in 94, human video in 54, web vision–language data in 45 and motion capture in 19 (a model can use several). Data scale is reported for 184 models but in incompatible units—episodes, trajectories, frames, hours—so we parsed hours only. Fifty-four models report hours of data (median 1,522 h). Sixteen report 10,000 h or more, and the largest figures come from 2025–2026 models: 270,000 h of real-world manipulation data for GEN-0 [52], over 500,000 h of data from human wearable devices for GEN-1 [53], and over one million hours of video for DYNA-2 (egocentric human video) [54] and V-JEPA 2-AC (internet video) [55]. Together with studies of data scaling in imitation learning [56] and large open datasets such as Open X-Embodiment [44] and DROID [57], these figures indicate that the field is now scaling data faster than parameters.

7. Disclosure Audit

Figure 6 shows the share of models that disclose each of the twelve items. What a model is made of is usually reported: the backbone (87%), the head type (87%) and the scale of the training data (86%). What it costs is not. Training accelerators are named for 47% of models, training duration for 27%, pretraining GPU-hours can be computed for 18% and inference memory is reported for 13%. Only 41% report a control rate and 51% name the inference device. Half (53%) report a total parameter count, the single number that the memory estimate in Section 8 needs most.

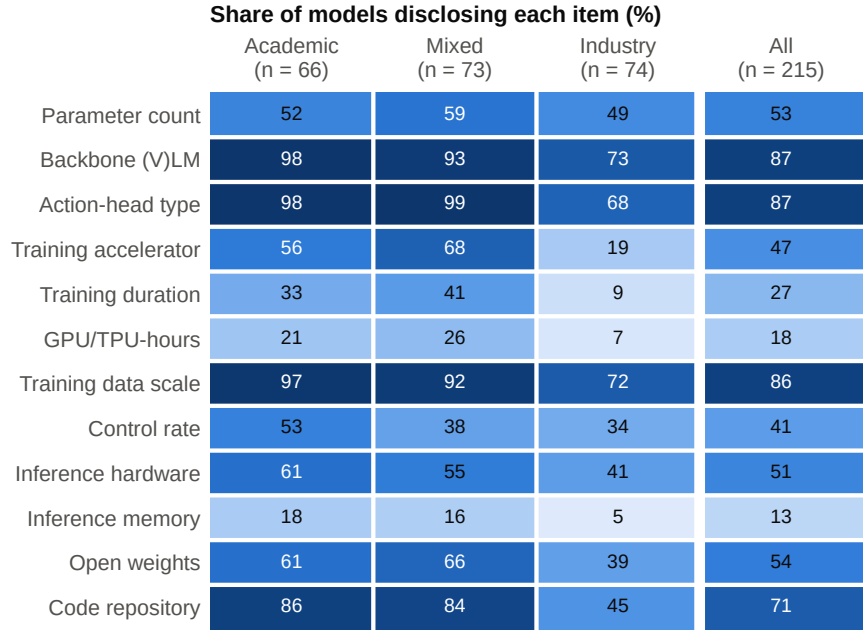


Figure 6. Share of models that disclose each resource item, by developer sector. Two models with unknown affiliation are counted only in "All".

Disclosure differs by sector (Table 3). Industry-only models disclose eight of the twelve items significantly less often than other models after Holm correction. The gaps are widest for training cost: 19% of industry-only models name a training accelerator against 63% of the others, 9% against 37% report a training duration, and 7% against 24% allow GPU-hours to be computed. Industry-only models also disclose the head type less often (68% against 99%), release weights less often (39% against 63%) and have an official repository less often (45% against 85%). The parameter count is the exception: about half of the models in every sector report it. Counting items, the median model discloses 6 of the 12; the medians are 7 for academic-only and mixed teams and 4.5 for industry-only teams.

Table 3. Disclosure of each item by industry-only models versus all other models (academic-only and mixed teams; two models with unknown affiliation excluded). OR: odds ratio from Fisher's exact test; p_{Holm}: Holm-adjusted p-value over the 12 items.

Item	Industry (n = 74)	Other (n = 139)	OR	p_{Holm}
Parameter count	36 (49%)	77 (55%)	0.76	0.39
Backbone (V)LM	54 (73%)	133 (96%)	0.12	<0.001
Action-head type	50 (68%)	137 (99%)	0.03	<0.001
Training accelerator	14 (19%)	87 (63%)	0.14	<0.001
Training duration	7 (9%)	52 (37%)	0.17	<0.001
GPU/TPU-hours	5 (7%)	33 (24%)	0.23	0.01
Training data scale	53 (72%)	131 (94%)	0.15	<0.001
Control rate	25 (34%)	63 (45%)	0.62	0.22
Inference hardware	30 (41%)	80 (58%)	0.50	0.07
Inference memory	4 (5%)	24 (17%)	0.27	0.07
Open weights	29 (39%)	88 (63%)	0.37	0.005
Code repository	33 (45%)	118 (85%)	0.14	<0.001

Disclosure is also falling over time. Between 2024 and 2026, the share of models that name a training accelerator fell from 57% to 31%, that report a training duration from 37% to 12% and that disclose the head type from 97% to 67%. Part of the 2026 drop may be temporary, because some 2026 releases have so far been described only in blog posts that may be followed by papers; but the gap between sectors is present in every year. The practical consequence is a selection bias: any comparison of training cost or inference speed across VLAs is, in effect, a comparison among academic and open models.

8. Estimating Deployment Feasibility

The disclosure audit shows that latency and memory are rarely reported for the device a reader cares about. We therefore estimate them from the few quantities that are usually available—total parameters, head type, chunk structure—and from device specifications, and we test the estimate against every published latency measurement in the corpus.

8.1. Memory

The inference memory of a model with P billion parameters stored at b bytes per parameter is estimated as

$$M = P\,b + 1\ \text{GB} + 0.3\ \text{GB} \times P, \tag{1}$$

where the constant term covers the runtime and the last term activations, key–value caches and the vision encoder's working memory. A model is taken to fit when M is at most 90% of device memory, or 75% on Jetson modules, whose memory is shared with the CPU. Quantised precisions ($b = 1$ for int8 and $b = 0.5$ for int4) assume that a working quantised implementation exists, as for NaVILA with AWQ [8,58].

8.2. Latency

Each network pass is modelled with a roofline [59]: it takes as long as the slower of reading the weights from memory and doing the arithmetic, plus a fixed overhead. For a network of p billion parameters processing n tokens on a device with memory bandwidth B_{mem} (GB/s) and dense BF16 throughput F_{bf16} (TFLOPS),

$$f(p,n) = \max\left(\frac{1000\,p\,b}{\beta B_{\text{mem}}},\ \frac{2\,p\,n}{\varphi F_{\text{bf16}}}\right) + c_0 \quad [\text{ms}], \tag{2}$$

where β and φ are the achievable fractions of peak bandwidth and compute and c_0 is a per-pass overhead. The latency of one inference, which produces one action or one action chunk, depends on the head:

$$L_{\text{tokens}} = f(P, n_{\text{in}}) + n_{\text{tok}}\left(\frac{1000\,P\,b}{\beta B_{\text{mem}}} + c_t\right), \tag{3}$$

$$L_{\text{iterative}} = f(P - P_h, n_{\text{in}}) + K f(P_h, 50), \tag{4}$$

$$L_{\text{single}} = f(P, n_{\text{in}}). \tag{5}$$

Token heads (Equation (3)) run one prefill pass over n_{in} input tokens and then decode n_{tok} action tokens, each of which re-reads the weights; this is the memory-bound regime of autoregressive decoding [60]. Flow-matching and diffusion heads (Equation (4)) run the backbone once and a head of P_h parameters K times over about 50 action tokens. Regression, CVAE and RL heads (Equation (5)) need one pass. On Jetson modules the result is multiplied by γ to account for lower sustained clocks and less mature kernels. The defaults are $n_{\text{in}} = 600$ input tokens for models with a VLM (1,500 for navigation models that encode video history) and 64 otherwise, $n_{\text{tok}} = 7$ (5 for navigation) unless the source states the number of tokens per action, $K = 10$ unless the source states the number of steps. When the head size is not reported, P_h is set to 0.3B for VLM-based models of at least 1.5B parameters, to $0.25P$ for smaller VLM-based models and to $0.7P$ for models without a VLM, whose head is most of the network. Token policies without a language model, such as RT-1 [37], emit all action tokens in one pass and are treated as single-pass. Video world-action models are not estimated because their cost depends on the video resolution and horizon.

The estimate needs a total parameter count. When only the language-model size P_{LM} is known, we use $P \approx 1.15\,P_{\text{LM}}$, plus P_h for iterative heads. This is possible for 128 of the 215 models (105 from the total count, 23 from the language-model size); the remaining 87 cannot be assessed.

8.3. Calibration and Validation

The corpus contains 38 published latency measurements ("anchors") for 30 models on 11 devices (Table A3 in Appendix B); 37 can be estimated, the exception being a video world-action model. Device specifications for 24 devices, from the 8 GB Jetson Orin Nano to data-centre GPUs, are listed in Table A2. The five constants were fitted to these anchors and rounded to $\beta = \varphi = 0.6$, $c_0 = c_t = 4$ ms and $\gamma = 1.5$; a finer grid search reaches a median error of 1.29× on the same anchors, so rounding costs little. Accuracy is measured as the fold error $\max(\hat{L}/L, L/\hat{L})$.

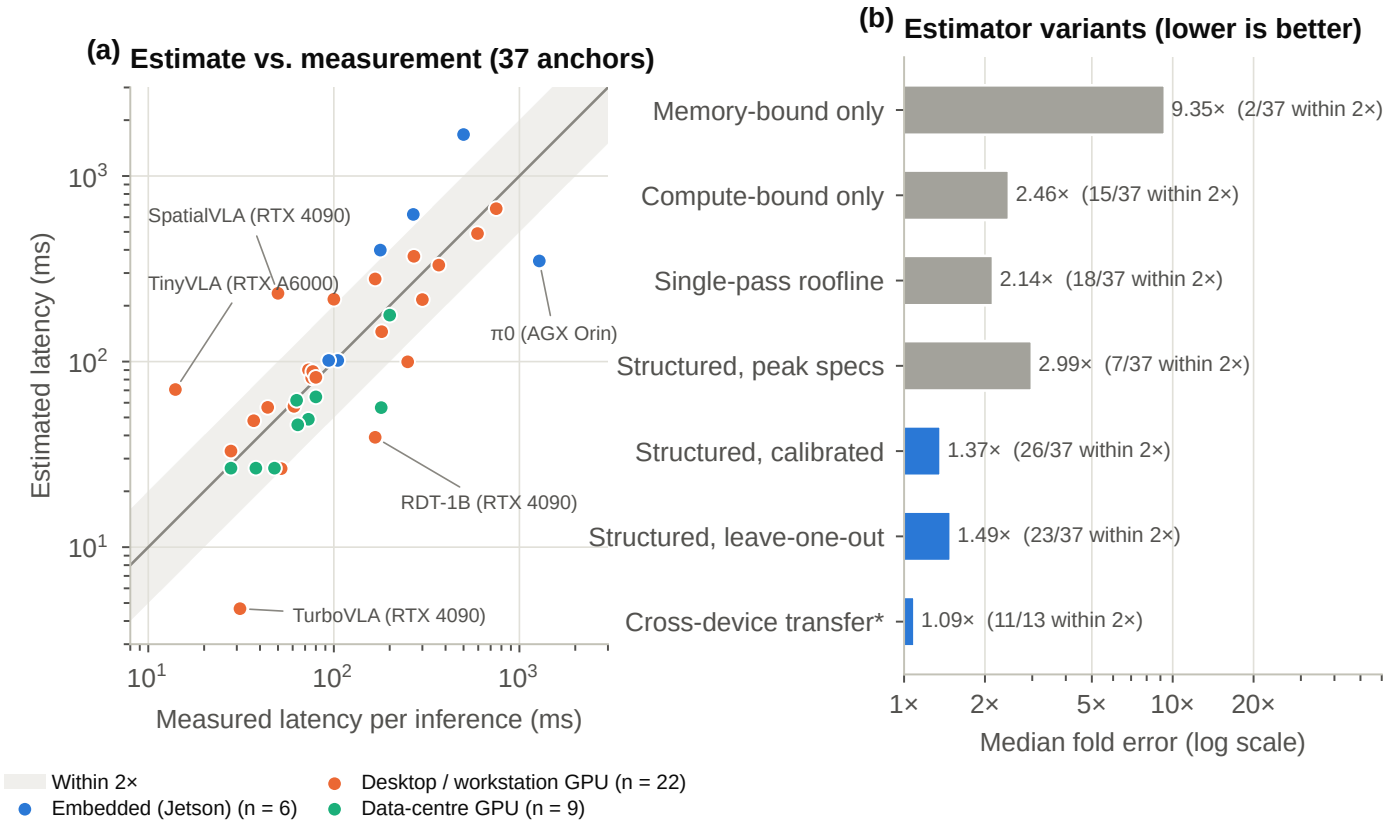


Figure 7. Validation of the latency estimator. (**a**) Estimated against measured latency for the 37 anchors; the shaded band marks agreement within a factor of two, and the five largest errors are labelled. (**b**) Median fold error of estimator variants. *Cross-device transfer predicts each anchor from another measurement of the same model ($n = 13$).

With the calibrated constants the median fold error is 1.37×, and 26 of the 37 anchors are within a factor of two (23 within 1.5×; Figure 7, Table 4). The estimate is nearly unbiased: the geometric mean of the estimated-to-measured ratio is 0.95. Because the constants were fitted to the same anchors, we also ran a leave-one-out test, refitting the constants on 36 anchors and predicting the held-out one. The out-of-sample median error is 1.49× with 23 of 37 within a factor of two, so the in-sample figure is only mildly optimistic.

Table 4. Accuracy of estimator variants on the published latency anchors. Fold error is $\max(\hat{L}/L, L/\hat{L})$; bias is the geometric mean of $\hat{L}/L$ (values below 1 mean underestimation). The first three variants use the calibrated efficiency constants; "peak specifications" uses $\beta = \varphi = \gamma = 1$ and no fixed overheads.

Variant	n	Median fold error	≤1.5×	≤2×	Bias
Memory-bound only (weights read once)	37	9.35×	0	2	0.09
Compute-bound only ($2Pn$ FLOPs)	37	2.46×	8	15	0.26
Single-pass roofline, no head structure	37	2.14×	10	18	0.39
Structured roofline, peak specifications	37	2.99×	7	7	0.28
Structured roofline, calibrated (this work)	37	1.37×	23	26	0.95
Structured roofline, leave-one-out	37	1.49×	19	23	0.87
Cross-device transfer from a measured anchor	13	1.09×	10	11	0.93

Each part of the model earns its place. Using peak specifications without calibration underestimates latency by a factor of about 3.6 (median error 2.99×, 7 of 37 within 2×). With calibrated constants, a memory-bound-only model (weights read once) errs by a median of 9.35×, a compute-bound-only model by 2.46× and a single-pass roofline that

ignores the head structure by 2.14×. Representing token decoding and iterative denoising explicitly is what brings the error down to 1.37×. This supports the qualitative claim of Grover et al. [17] that autoregressive VLAs are bandwidth-bound and diffusion-style heads compute-bound, and puts numbers on it.

Errors are smallest on desktop and data-centre GPUs (median 1.30× on 19 desktop anchors and 1.40× on 9 data-centre anchors) and largest on Jetson modules (2.28× on 6 anchors), where power modes, unified memory and software maturity vary most. The five largest errors fall into three groups. For very small models such as TurboVLA (0.2B) [61], fixed software overheads dominate and the estimate understates latency 6.7-fold. For TinyVLA [62] and SpatialVLA [63], the head structure or the number of decoded tokens that we had to assume overstates the work, and the estimate overstates latency about fivefold. For RDT-1B [41] and for π_0 on a Jetson AGX Orin [3], the measurement probably includes work that the estimate does not model, such as separate text or image encoders, preprocessing or a slower power mode, and the estimate understates latency about fourfold.

When a model has been measured on one device, the estimator can transfer that measurement to another device by scaling it with the ratio of the two estimates. On the 13 anchors that have a second anchor of the same model, this transfer has a median error of only 1.09×, with 11 of 13 within a factor of two. The gallery therefore uses a measured value when one exists for the same device and precision, a transferred value when the model was measured on another device, and the pure estimate otherwise.

8.4. Which Models Run Where

Figure 8 applies the estimator to the 126 models that can be assessed on eleven representative devices. A model counts as deployable if it fits in memory and the estimated rate, or the measured or transferred rate when available, is at least 10 Hz. Three regimes appear. On 8–16 GB Jetson modules memory is the binding constraint: in bf16, 89 of the 126 models do not fit in the 8 GB of an Orin Nano and 54 do not fit in the 16 GB of an Orin NX, and only 19 and 20 models reach 10 Hz. On the 64 GB AGX Orin almost everything fits (six models do not) but most models are too slow: 24 reach 10 Hz and 95 run but below 10 Hz. On desktop and data-centre GPUs both constraints relax, and 68 models reach 10 Hz on an RTX 4090, 78 on an RTX 5090 and 105 on an H100.

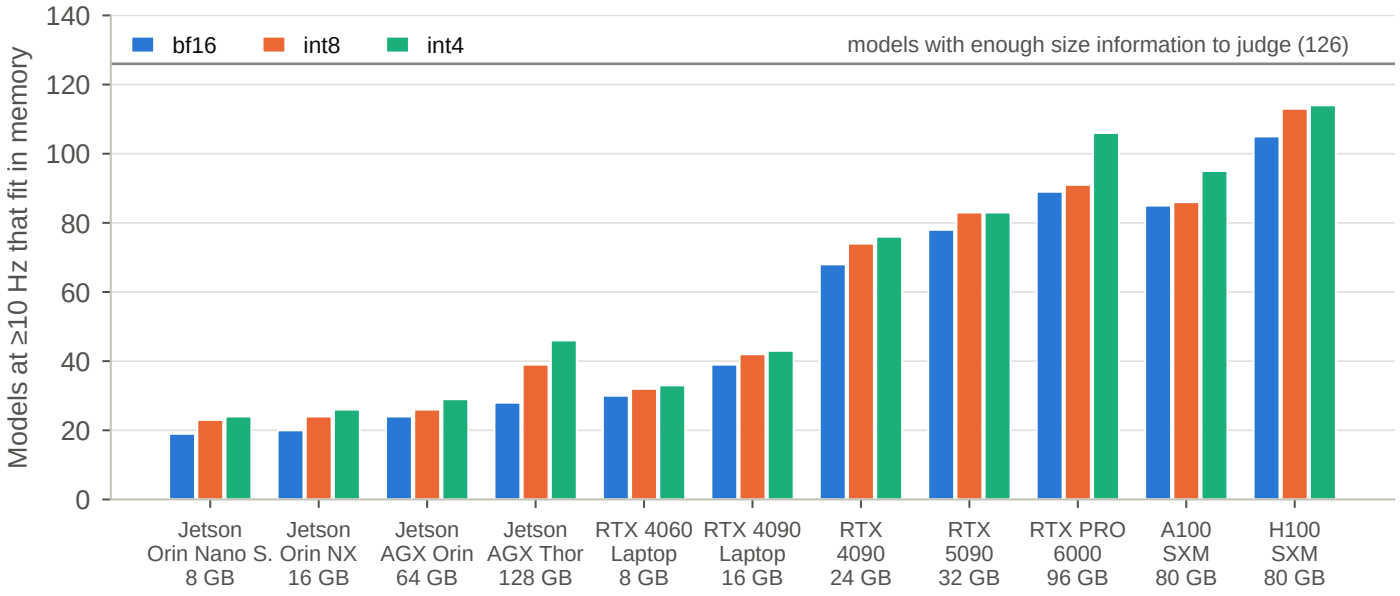


Figure 8. Number of models estimated to fit in memory and run at 10 Hz or faster on eleven devices, by inference precision. The horizontal line marks the 126 models with enough size information to be assessed.

Quantisation helps most where decoding dominates. On the Jetson AGX Thor, moving from bf16 to int4 raises the number of models at 10 Hz from 28 to 46, and at 5 Hz from 66 to 84, because token-decoding models become faster when fewer bytes are read. On the Orin Nano, int4 lets 45 more models fit in memory, but most of them are too slow, so the number at 10 Hz rises only from 19 to 24. The largest models that reach 10 Hz on an Orin Nano in bf16 are TinyVLA (1.3B) [62] and HPT (1.1B) [64]; on an RTX 4090 they include 7B models such as TrackVLA [65] and 5B models such as $\pi_{0.7}$ [66]. These are estimates for batch size one without compiler or kernel optimisation; optimised runtimes such as TensorRT, which GR00T N1.7 uses for its reported latencies [51], can change measured rates substantially.

8.5. Fine-Tuning Cost

A similar estimate covers fine-tuning. Per-GPU memory is the sum of model state and activations. Full fine-tuning with mixed-precision Adam needs about 16 bytes per parameter (bf16 weights and gradients, an fp32 master copy and two optimiser moments [67]), divided by the number of GPUs when the state is sharded [67,68]. LoRA [69] keeps frozen bf16 weights (2 bytes per parameter) and trains adapters with 1% of the parameters; QLoRA [70] stores the frozen weights in 4 bits (about 0.55 bytes per parameter); training only the action head keeps the backbone frozen in bf16. Activations are modelled as 2 GB plus 1.5 GB per billion parameters for small batches and 4 GB plus 5 GB per billion for typical batches. These activation constants were set to match three reported requirements—more than 22.5 GB for LoRA and more than 70 GB for full fine-tuning of π_0, about 27 GB for LoRA on OpenVLA at batch size one, and about 70 GB for full fine-tuning of GO-1 [2,3,71]—and, unlike the latency model, have not been validated on independent measurements. Training time follows the usual $6PN$ rule [72] ($4PN$ for adapter training, $2PN$ for a frozen backbone) at 35% of peak BF16 throughput and 90% scaling efficiency across GPUs.

Table 5. Estimated fine-tuning feasibility for the 86 open-weight models with enough size information. Each cell gives the number of models whose estimated per-GPU memory fits (90% of device memory; 75% on Jetson) and, in parentheses, the largest such model (total parameters). Memory assumes the "normal" activation budget (4 GB + 5 GB per billion parameters per GPU). Head-only applies to the models with a separable flow-matching or diffusion head of known size.

Configuration	Full	LoRA	QLoRA	Head only
1× RTX 4090 (24 GB)	19 (≤0.77B)	30 (≤2.4B)	44 (≤3B)	33 (≤5.1B)
1× RTX 5090 (32 GB)	24 (≤1.1B)	46 (≤3.3B)	53 (≤4.2B)	37 (≤7.6B)
1× A100 (80 GB)	45 (≤3.2B)	83 (≤9B)	83 (≤9B)	42 (≤8.4B)
8× H100 (80 GB)	83 (≤9B)	83 (≤9B)	83 (≤9B)	42 (≤8.4B)
Jetson AGX Thor (128 GB)	53 (≤4.2B)	83 (≤9B)	85 (≤14B)	42 (≤8.4B)

Table 5 shows what this implies for the 86 open-weight models that can be assessed. A single 24 GB RTX 4090 can fully fine-tune only models below about 0.8B parameters (19 models), but LoRA reaches 2.4B (30 models) and QLoRA 3B (44 models). One 80 GB A100 is enough for LoRA on 83 of the 86 models, up to 9B; the remaining three are the largest in the set (two 14B models and a system whose planner is a 72B VLM). For example, the estimate for LoRA on OpenVLA (7B) is about 54 GB at a typical batch size, so it needs an 80 GB GPU, and 10,000 steps at a global batch of 32 take an estimated 14 h on one A100; on eight H100 GPUs, full fine-tuning of π_0 takes an estimated 23 min.

9. Reinforcement Learning in VLA Pipelines

Most models in the corpus are trained by imitation alone: 132 do not use RL, 31 use it as a core component, 28 use it partially and 24 do not say. Where RL appears, it enters at one of

six points, and the point depends strongly on the domain (Figure 9a). In humanoid models RL trains the controller: 13 humanoid models are RL-trained whole-body controllers, often motion trackers learned in simulation [50,73,74], and 12 place an RL low-level controller under a VLA that issues commands or latent goals [75]. Mobile manipulation follows the same pattern on legged robots, with RL-trained controllers in five models and RL low-level controllers in four [76,77].

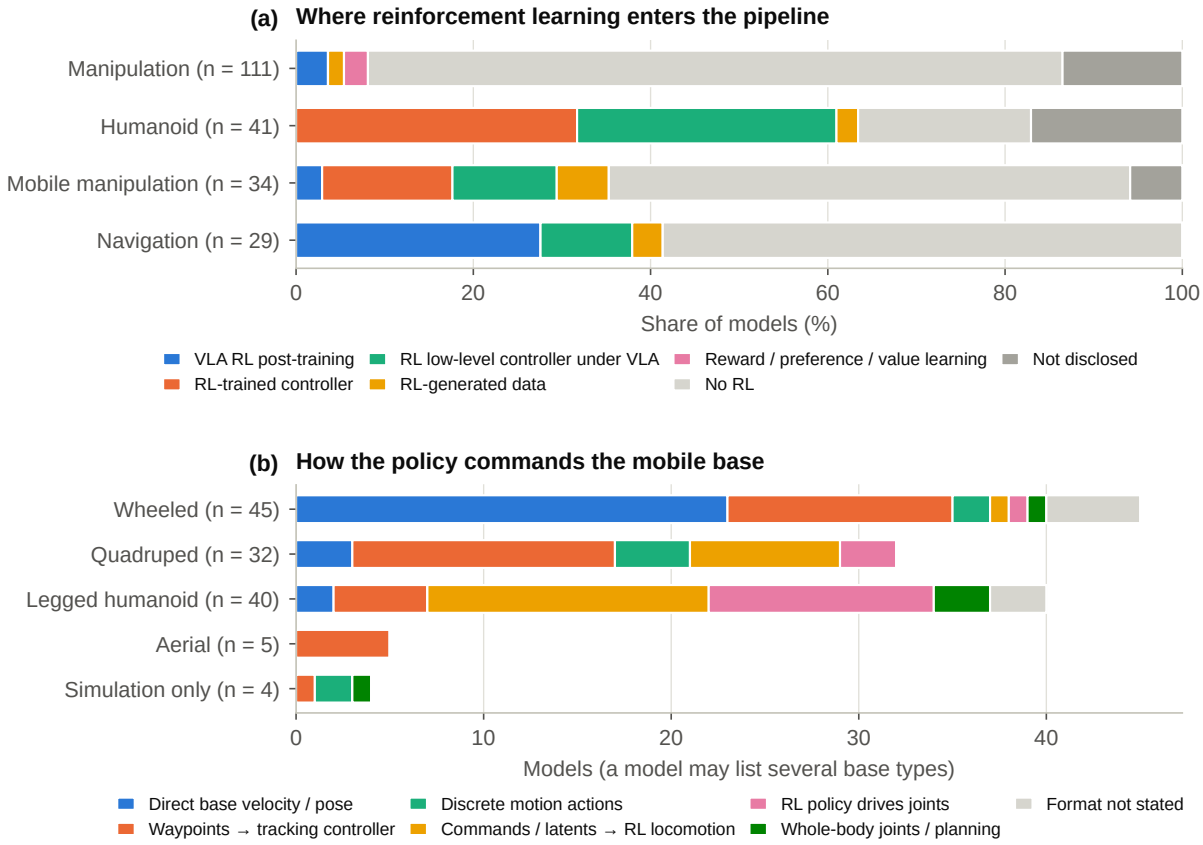


Figure 9. (**a**) Where reinforcement learning enters the pipeline, by domain. (**b**) How the policy commands the mobile base, by base type; a model can list more than one base type.

RL post-training of the VLA itself is the newest use and is concentrated in navigation. Of the 13 models that fine-tune the VLA with RL, eight are navigation models [78–85]. They train in simulation, where rewards such as progress towards a goal are cheap to compute, using PPO-style objectives [86] or group-relative policy optimisation (GRPO) [87]. In manipulation, RL post-training is rarer (four models) and relies on real-robot or offline data: $\pi^*_{0.6}$ learns a value function from demonstrations, autonomous rollouts and human corrections and uses it to improve the policy [88], GR-RL runs online RL on a real robot [89], and SyVLA optimises the whole VLA with a PPO variant on a real robot [90]. One quadruped model applies offline RL to its VLA [91]. Three manipulation models use reward, preference or value learning without full RL post-training, and six models across domains use data generated by RL policies. Because simulation RL runs on one or a few GPUs (Section 6), adding an RL controller changes the training budget far less than VLA pretraining does.

10. Commanding the Mobile Base

98 models control a robot with a mobile base: 45 wheeled, 32 quadruped, 40 legged humanoid, five aerial and four simulation-only platforms (a model can list several). How the policy talks to the base is rarely stated in one place, so we coded it into eight classes

(Figure 9b). The interface follows the platform. On wheeled robots the policy usually outputs base velocities or poses directly (23 models), as RT-1 does [37], or waypoints for a tracking controller (12), as in the navigation foundation models GNM, ViNT and NoMaD [92–94]. On quadrupeds waypoints dominate (14 models), followed by commands or latents for an RL locomotion controller (8). On humanoids the policy almost never drives the base directly: 15 models send commands or latent goals to an RL locomotion controller and 12 are themselves RL policies that drive the joints. All five aerial models output waypoints. Eight models do not state the interface at all.

These choices matter for resources. A policy that outputs waypoints or latent commands can run at a few hertz while a separately trained controller closes the loop at hundreds of hertz, which is how large VLMs are used on legged robots despite the latencies in Section 8. The cost is that the controller, its training and its rate become part of the system that must be reported.

11. Discussion

11.1. Implications for Practitioners

Three practical lessons follow. First, check memory before speed on small embedded computers and speed before memory on large ones. Most of the assessable models do not fit in bf16 on an 8 GB Orin Nano and almost half do not fit on a 16 GB Orin NX, so quantisation is the first lever there; on 64–128 GB modules nearly everything fits, and the head structure and chunking decide whether the policy is fast enough. Second, the head structure matters as much as the parameter count. A 3B model with a flow-matching head and 50-step chunks can control a robot at 50 Hz on a desktop GPU [3], whereas a 7B model that decodes several tokens per action is limited by memory bandwidth to a few hertz [2] unless it uses parallel decoding and chunking [95]. Third, fine-tuning is within reach of a single workstation for most open models below about 3B parameters if LoRA or QLoRA is used, whereas full fine-tuning of 7B-class models still needs several 80 GB GPUs. The gallery that accompanies this review performs these checks for any of the 24 devices in Table A2 or for user-supplied specifications.

11.2. A Reporting Checklist

The disclosure audit points to a small set of numbers that, if reported consistently, would make the estimates above unnecessary. Table 6 lists ten items, in the spirit of model cards [96], datasheets [97], reproducibility checklists [98] and calls to report compute and energy [99–101]. Most cost the authors little: they are already known when the paper is written.

Table 6. A ten-item checklist for reporting the resources of robot foundation policies.

#	Item	Why it matters
1	Total parameters, split into vision encoder, backbone and action head, with the precision used at inference	Sets memory and the bandwidth-bound part of latency; reported by only 53% of models
2	Action representation: head type, chunk length, tokens per action or denoising steps	Determines how latency scales with model size (Equations (3)–(5))
3	For each training stage: accelerator model, count, wall-clock duration, steps and batch size	Allows GPU-hours to be computed; only 18% of models allow this today
4	Training data by source type (teleoperation, simulation, human video, web), in hours and episodes	Data volume, more than parameter count, tracks training cost (Section 6)
5	Fine-tuning recipe: method (full, LoRA, head only), GPUs, memory, steps and time	Tells users whether adaptation fits their hardware
6	Inference latency per action chunk on a named device, with precision, batch size and software stack	Enables like-for-like comparison and calibration of estimators
7	Peak inference memory on that device	Reported by only 13% of models
8	Control rate on the robot and where inference runs (onboard, local PC, remote server), including network latency	Separates model speed from system design
9	Interface to the robot: what the policy outputs, at what rate, and to which low-level controller	Required to reproduce mobile and humanoid systems (Section 10)
10	Release status of weights, code and training data, with licences, and any RL use with its simulation budget	Determines whether the model can be studied or deployed by others

Standard benchmarks of inference speed for robot policies, analogous to MLPerf for data-centre inference [102], would complement such reporting. The anchors in Table A3 show the problem: they were measured by different authors with different software stacks and sometimes include image preprocessing or network transfer, which limits how precisely any estimator can be calibrated.

11.3. Limitations

The review has several limitations. The corpus is large but not exhaustive; it reflects the search described in Section 3 and the reference dates of 6–19 October 2026, and the field moves fast enough that some 2026 models will have published papers after the cut-off. All values are self-reports by the developers; we did not reproduce training runs or measure latencies ourselves, and 12 records could not be fully verified. Coding the head family, the RL category, the base interface and the developer sector involves judgement, and inter-rater agreement was not measured. GPU-hours mix accelerator generations and exclude experimentation, so they understate true cost and should not be compared across generations without care.

The estimator is deliberately simple. It assumes batch size one, ignores compiler and kernel optimisation, assumes that quantised kernels exist, and treats a dual-system pipeline as a single network, which can make it optimistic for hierarchical models such as Helix whose large VLM runs at a lower rate than the action module. It needs a parameter count, so 87 models cannot be assessed, and it was validated on only 37 anchors, six of them on embedded devices. The fine-tuning estimates rest on three reported memory requirements and have not been validated independently. The estimates should therefore be read as screening tools, accurate to within a factor of two for about two thirds of the anchors, not as substitutes for measurement.

11.4. Future Work

Three extensions would strengthen this line of work: a shared protocol and open harness for measuring VLA latency, memory and energy on common devices, which would multiply the number and quality of anchors; adding energy and power to the coded resources, which matters for battery-powered robots; and keeping the dataset current, which the open gallery and scripts make possible.

12. Conclusions

We reviewed 215 robot policy models from October 2022 to October 2026 from the perspective of the resources they need to be trained and deployed. The architecture has moved towards flow-matching heads, action chunking and Qwen-family backbones of about 2–8B parameters, and model size has plateaued at a median of about 4B parameters since 2025, while training data have grown to hundreds of thousands of hours. Training cost spans more than three orders of magnitude, from tens to tens of thousands of GPU-hours, but it is reported for fewer than one model in five, and industry-only models disclose training hardware, training duration and compute far less often than academic and mixed teams. A roofline estimator that represents token decoding and iterative denoising predicts published latencies to a median factor of 1.37 (1.49 out of sample) and shows that memory limits deployment on small embedded computers while speed limits it on larger ones. We hope that the dataset, the estimator and the reporting checklist will make resource numbers a routine part of how robot foundation models are described.

Author Contributions: Conceptualization, F.L.; methodology, F.L.; software, F.L.; validation, F.L.; formal analysis, F.L.; investigation, F.L.; data curation, F.L.; writing—original draft preparation, F.L.; writing—review and editing, F.L.; visualization, F.L. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The dataset (215 model records, training-hardware records, device specifications and latency anchors), the analysis scripts that produce every number and figure in this article, and an interactive gallery are openly available at https://github.com/yc424k/vla-gallery and https://vla-gallery.is-a.dev/.

Acknowledgments: During the preparation of this manuscript, the author used Claude (Anthropic) [TODO(author): version] for the purposes of writing analysis and plotting code and drafting the manuscript text [TODO(author): and, if applicable, assisting data extraction and verification]. The author has reviewed and edited the output and takes full responsibility for the content of this publication.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BF16	Brain floating point, 16 bits
CVAE	Conditional variational autoencoder
FSDP	Fully sharded data parallel
GPU	Graphics processing unit
GRPO	Group relative policy optimisation
LLM	Large language model
LM	Language model
LoRA	Low-rank adaptation
QLoRA	Quantised low-rank adaptation
RL	Reinforcement learning
TPU	Tensor processing unit
VA	Vision–action (visuomotor policy without a pretrained VLM)
VLA	Vision–language–action
VLM	Vision–language model

Appendix A. The Corpus

Table A1 lists every model with its key attributes and primary source. Full records, including the source of every value, are in the released dataset.

Table A1. All 215 models in the corpus, ordered by release month. Domain: Manip. = manipulation, Hum. = humanoid, MoMa = mobile manipulation, Nav. = navigation. Params: total parameters in billions (n/d = not disclosed). LM: language-model family of the backbone. Head: action-head family. W: weights released (Y = yes, P = partially, N = no, ? = unverified). GPU-h: pretraining GPU/TPU-hours where both accelerator count and duration are reported.

#	Model	Date	Domain	Type	Params	LM	Head	W	GPU-h
1	GNM [92]	2022-10	Nav.	VA	0.0087	none	regression	Y	
2	RT-1 [37]	2022-12	MoMa	VA	0.035	none	tokens	Y	
3	Diffusion Policy (CNN) [31]	2023-03	Manip.	VA	0.309	none	diffusion	Y	
4	Diffusion Policy (Transformer) [31]	2023-03	Manip.	VA	0.102	none	diffusion	Y	
5	ACT [30]	2023-04	Manip.	VA	0.08	none	CVAE	N	
6	ViNT [93]	2023-06	Nav.	VA	0.031	none	regression	Y	240
7	RT-2 [1]	2023-07	MoMa	VLA	55	PaLI/PaLM	tokens	N	
8	NoMaD [94]	2023-10	Nav.	VA	0.019	none	diffusion	Y	
9	RT-1-X [44]	2023-10	Manip.	VA	0.035	none	tokens	Y	
10	RT-2-X [44]	2023-10	Manip.	VLA	55	PaLI/PaLM	tokens	N	
11	RoboFlamingo [103]	2023-11	Manip.	VLA	9	other	regression	Y	
12	GR-1 [104]	2023-12	Manip.	VLA	0.195	none	regression	P	
13	HarmonicMM [105]	2023-12	MoMa	VA	n/d	none	RL policy	N	
14	Octo [47]	2023-12	Manip.	VA	0.093	none	diffusion	Y	1,792
15	QUAR-VLA (QUART) [106]	2023-12	MoMa	VLA	8	other	tokens	N	
16	SARA-RT [107]	2023-12	Manip.	Other	n/d	PaLI/PaLM	tokens	N	
17	AutoRT [36]	2024-01	MoMa	Other	n/d	n/d	other	N	
18	Mobile ALOHA [108]	2024-01	MoMa	VA	n/d	none	CVAE	N	
19	Humanoid Next-Token Locomotion [109]	2024-02	Hum.	VA	0.008	none	regression	?	
20	NaVid [110]	2024-02	Nav.	VLA	7	Llama	tokens	Y	672
21	UMI [111]	2024-02	Manip.	VA	n/d	none	diffusion	P	
22	3D-VLA [112]	2024-03	Manip.	VLA	n/d	other	tokens	P	
23	DP3 [113]	2024-03	Manip.	VA	n/d	none	diffusion	N	
24	Visual Quadrupedal Loco-Manipulation [114]	2024-03	MoMa	VA	n/d	none	diffusion	N	
25	RFM-1 [115]	2024-03	Manip.	VLA	8	n/d	n/d	N	
26	RoboDuet [77]	2024-03	MoMa	Other	n/d	none	RL policy	N	
27	RT-H [45]	2024-03	MoMa	VLA	55	PaLI/PaLM	tokens	N	
28	VBC [76]	2024-03	MoMa	VA	n/d	none	RL policy	P	
29	VQ-BeT [116]	2024-03	Manip.	VA	n/d	none	tokens	P	
30	RISE [117]	2024-04	Manip.	VA	n/d	none	diffusion	N	
31	Tesla Optimus [118]	2024-05	Hum.	Other	n/d	n/d	n/d	N	
32	BAKU [119]	2024-06	Manip.	VA	0.114	none	regression	N	
33	HumanPlus HIT [120]	2024-06	Hum.	VA	n/d	none	regression	?	
34	HumanPlus HST [120]	2024-06	Hum.	VA	n/d	none	RL policy	?	
35	LLaRA [121]	2024-06	Manip.	VLA	7	Llama	tokens	Y	
36	OmniH2O [122]	2024-06	Hum.	VA	n/d	none	RL policy	?	

#	Model	Date	Domain	Type	Params	LM	Head	W	GPU-h
37	OpenVLA [2]	2024-06	Manip.	VLA	7	Llama	tokens	Y	21,500
38	RoboMamba [123]	2024-06	Manip.	VLA	2.8	Mamba	regression	?	
39	ECoT [124]	2024-07	Manip.	VLA	7	Llama	tokens	Y	
40	Mobility VLA [125]	2024-07	Nav.	VLA	n/d	Gemini	other	N	
41	UMI on Legs [126]	2024-07	MoMa	VA	n/d	none	diffusion	P	
42	CrossFormer [127]	2024-08	Nav.	VLA	0.13	none	regression	Y	12,032
43	Helpful DoggyBot [128]	2024-09	MoMa	Other	n/d	none	RL policy	N	
44	HPT [64]	2024-09	Manip.	VA	1.1	none	regression	P	
45	InterACT [129]	2024-09	Manip.	VA	n/d	none	regression	?	
46	Robot Utility Models (RUM) [130]	2024-09	MoMa	VA	n/d	none	tokens	Y	
47	ScaleDP [131]	2024-09	Manip.	VA	1	none	diffusion	?	
48	TinyVLA [62]	2024-09	Manip.	VLA	1.3	other	diffusion	P	
49	ALOHA Unleashed [132]	2024-10	Manip.	VA	0.217	none	diffusion	?	
50	Challenging-Terrain Transformer [133]	2024-10	Hum.	VA	0.0014	none	RL policy	?	
51	Data Scaling Laws [56]	2024-10	Manip.	VA	0.396	none	diffusion	Y	
52	Dita [134]	2024-10	Manip.	VLA	0.334	none	diffusion	Y	
53	EgoMimic [135]	2024-10	Manip.	VA	n/d	none	regression	?	
54	GR-2 [136]	2024-10	Manip.	VLA	0.23	none	CVAE	?	
55	HOVER [50]	2024-10	Hum.	VA	n/d	none	RL policy	?	24
56	iDP3 [137]	2024-10	Hum.	VA	n/d	none	diffusion	Y	
57	LAPA [138]	2024-10	Manip.	VLA	7	Llama	latent	Y	272
58	LeLaN [139]	2024-10	Nav.	VLA	n/d	none	regression	Y	
59	RDT-1B [41]	2024-10	Manip.	VLA	1.2	none	diffusion	Y	34,560
60	X-Mobility [140]	2024-10	Nav.	VA	n/d	none	regression	Y	
61	π_0 [3]	2024-10	Manip.	VLA	3.3	Gemma	flow	Y	
62	CityWalker [141]	2024-11	Nav.	VA	0.214	none	regression	Y	60
63	CogACT [142]	2024-11	Manip.	VLA	7.6	Llama	diffusion	Y	1,920
64	TidyBot++ (diffusion policy) [143]	2024-11	MoMa	VA	n/d	none	diffusion	P	
65	CARP [144]	2024-12	Manip.	VA	0.0161	none	tokens	Y	
66	DiVLA (Diffusion-VLA) [46]	2024-12	Manip.	VLA	72	Qwen	diffusion	?	
67	ExBody2 [145]	2024-12	Hum.	VA	n/d	none	RL policy	?	
68	MiniVLA [146]	2024-12	Manip.	VLA	1	Qwen	tokens	Y	
69	Moto [147]	2024-12	Manip.	VA	0.098	none	latent	Y	
70	NaVILA [8]	2024-12	Nav.	VLA	8	Llama	tokens	Y	
71	NWM (Navigation World Models) [148]	2024-12	Nav.	Other	1	none	video WM	Y	
72	QUART-Online [149]	2024-12	MoMa	VLA	8	other	tokens	Y	
73	RING (The One RING) [79]	2024-12	Nav.	VLA	n/d	none	tokens	Y	
74	TraceVLA [150]	2024-12	Manip.	VLA	7	Llama	tokens	Y	
75	UH-1 (Humanoid-X) [151]	2024-12	Hum.	Other	n/d	none	tokens	?	440
76	Uni-NaVid [9]	2024-12	Nav.	VLA	7	Llama	tokens	Y	1,400
77	SLIM [152]	2025-01	MoMa	VLA	0.019	none	RL policy	N	
78	SpatialVLA [63]	2025-01	Manip.	VLA	3.5	Gemma	tokens	Y	15,360
79	π_0-FAST [153]	2025-01	Manip.	VLA	3	Gemma	tokens	Y	
80	COMPASS [78]	2025-02	Nav.	VA	n/d	none	RL policy	P	
81	DexGraspVLA [154]	2025-02	Manip.	VLA	n/d	Qwen	diffusion	Y	
82	DexVLA [155]	2025-02	Manip.	VLA	n/d	Qwen	diffusion	P	
83	Figure Helix [5]	2025-02	Hum.	VLA	7.1	n/d	regression	N	
84	Hi Robot [156]	2025-02	Manip.	VLA	n/d	Gemma	flow	N	
85	Humanoid-VLA [157]	2025-02	Hum.	VLA	n/d	Llama	tokens	?	1,728
86	Magma [158]	2025-02	Manip.	VLA	8.6	Llama	tokens	Y	
87	OpenVLA-OFT [95]	2025-02	Manip.	VLA	7	Llama	regression	Y	
88	BRS (WB-VIMA) [159]	2025-03	MoMa	VA	n/d	none	diffusion	?	
89	CoT-VLA [160]	2025-03	Manip.	VLA	7	Llama	tokens	?	11,000
90	Dense Policy [161]	2025-03	Manip.	VA	n/d	none	regression	N	
91	Figure RL walking controller [162]	2025-03	Hum.	VA	n/d	none	RL policy	N	
92	Gemini Robotics [34]	2025-03	Manip.	VLA	n/d	Gemini	n/d	N	
93	Gemini Robotics-ER [34]	2025-03	Manip.	Other	n/d	Gemini	other	N	
94	GO-1 (Genie Operator-1) [71]	2025-03	MoMa	VLA	3	InternLM	diffusion	Y	
95	GR00T N1 [4]	2025-03	Hum.	VLA	2.2	SmolLM	flow	Y	50,000
96	HAT [163]	2025-03	Hum.	VA	n/d	none	CVAE	?	
97	HybridVLA [164]	2025-03	Manip.	VLA	7	Llama	hybrid	Y	10,000
98	MoManipVLA [7]	2025-03	MoMa	VLA	7	Llama	tokens	N	
99	MoRE [91]	2025-03	MoMa	VLA	9.8	other	tokens	N	800
100	NORA [165]	2025-04	Manip.	VLA	3	Qwen	tokens	Y	4,000
101	$\pi_{0.5}$ [6]	2025-04	MoMa	VLA	n/d	Gemma	flow	P	
102	AMO [166]	2025-05	Hum.	VA	n/d	none	RL policy	P	
103	GR00T N1.5 [167]	2025-05	Hum.	VLA	3	n/d	flow	Y	
104	GraspVLA [168]	2025-05	Manip.	VLA	n/d	InternLM	flow	Y	
105	NavDP [169]	2025-05	Nav.	VA	n/d	none	diffusion	Y	
106	TrackVLA [65]	2025-05	Nav.	VLA	7	Llama	diffusion	N	360
107	TWIST [170]	2025-05	Hum.	VA	n/d	none	RL policy	Y	

#	Model	Date	Domain	Type	Params	LM	Head	W	GPU-h
108	UniVLA (OpenDriveLab) [171]	2025-05	Manip.	VLA	7	Llama	latent	Y	960
109	$\pi_{0.5}$ + KI [172]	2025-05	Manip.	VLA	n/d	Gemma	hybrid	P	
110	1X Redwood AI [173]	2025-06	Hum.	VLA	0.16	none	diffusion	N	
111	BumbleBee [174]	2025-06	Hum.	VA	n/d	n/d	n/d	?	
112	Chain-of-Action [175]	2025-06	Manip.	VA	n/d	none	regression	?	
113	DYNA-1 [176]	2025-06	Manip.	Other	n/d	n/d	n/d	N	
114	Gemini Robotics On-Device [177]	2025-06	Manip.	VLA	n/d	Gemini	n/d	N	
115	GMT (General Motion Tracking) [178]	2025-06	Hum.	VA	n/d	none	RL policy	Y	96
116	LeVERB [75]	2025-06	Hum.	VLA	n/d	none	latent	?	
117	LeVERB-A (LeVERB System 1) [75]	2025-06	Hum.	VA	n/d	none	RL policy	?	
118	Real-Time Chunking (inference-time) [35]	2025-06	Manip.	Other	n/d	Gemma	flow	N	
119	SmolVLA [179]	2025-06	Manip.	VLA	2.2	SmolLM	flow	Y	
120	UniVLA (BAAI) [180]	2025-06	Manip.	VLA	8.5	other	tokens	Y	
121	V-JEPA 2-AC [55]	2025-06	Manip.	VA	1.2	none	other	Y	
122	VLN-R1 [80]	2025-06	Nav.	VLA	7	Qwen	tokens	N	288
123	WorldVLA [181]	2025-06	Manip.	VLA	n/d	other	tokens	Y	
124	AC-DiT [182]	2025-07	MoMa	VLA	n/d	none	diffusion	N	
125	Being-H0 [183]	2025-07	Hum.	VLA	14	n/d	tokens	Y	
126	EgoVLA [184]	2025-07	Hum.	VLA	n/d	n/d	regression	?	
127	GR-3 [185]	2025-07	MoMa	VLA	4	Qwen	flow	?	
128	InternVLA-N1 [186]	2025-07	Nav.	VLA	8.4	Qwen	diffusion	Y	
129	Skild Brain [187]	2025-07	MoMa	Other	n/d	n/d	n/d	N	
130	StreamVLN [188]	2025-07	Nav.	VLA	8	Qwen	tokens	Y	1,500
131	TRI LBM 1.0 [189]	2025-07	Manip.	VLA	n/d	none	diffusion	?	
132	villa-X [190]	2025-07	Manip.	VLA	n/d	n/d	flow	P	6,144
133	Atlas LBM [191]	2025-08	Hum.	VLA	0.45	none	flow	N	
134	BeyondMimic [73]	2025-08	Hum.	VA	n/d	none	diffusion	?	
135	EO-1 (EmbodiedOneVision) [192]	2025-08	Manip.	VLA	3	Qwen	hybrid	Y	
136	Galaxea G0 [193]	2025-08	MoMa	VLA	n/d	Gemma	n/d	Y	
137	Genie Envisioner (GE-Base/GE-Act/GE-Sim) [194]	2025-08	Manip.	VLA	n/d	none	video WM	Y	8,832
138	MolmoAct [195]	2025-08	Manip.	VLA	7	Qwen	tokens	Y	9,728
139	RynnVLA-001 [196]	2025-08	Manip.	VLA	7	other	latent	Y	
140	F1-VLA [197]	2025-09	Manip.	VLA	4.2	Gemma	flow	Y	
141	FLOWER [198]	2025-09	Manip.	VLA	0.947	none	flow	Y	200
142	Gemini Robotics 1.5 [199]	2025-09	Manip.	VLA	n/d	Gemini	n/d	N	
143	Gemini Robotics-ER 1.5 [199]	2025-09	Manip.	Other	n/d	Gemini	other	N	
144	GR00T N1.6 [200]	2025-09	Hum.	VLA	3	n/d	flow	Y	
145	NavFoM [201]	2025-09	Nav.	VLA	7	Qwen	regression	N	4,032
146	OmniVLA [202]	2025-09	Nav.	VLA	7.5	Llama	regression	Y	
147	RDT2 (RDT2-FM / RDT2-VQ) [203]	2025-09	Manip.	VLA	n/d	Qwen	flow	Y	
148	UnifoLM-WMA-0 [204]	2025-09	Manip.	Other	n/d	n/d	video WM	Y	
149	VLA-Adapter [205]	2025-09	Manip.	VLA	n/d	Qwen	regression	Y	
150	WALL-OSS [206]	2025-09	MoMa	VLA	4	Qwen	hybrid	Y	
151	GigaBrain-0 [207]	2025-10	MoMa	VLA	n/d	Gemma	flow	Y	
152	InternVLA-M1 [208]	2025-10	Manip.	VLA	4.1	Qwen	diffusion	Y	
153	VLA-0 [209]	2025-10	Manip.	VLA	3	Qwen	tokens	Y	
154	X-VLA [210]	2025-10	Manip.	VLA	0.9	none	flow	Y	6,144
155	ACT-1 [211]	2025-11	MoMa	VA	n/d	n/d	n/d	N	
156	Evo-1 [212]	2025-11	Manip.	VLA	0.77	Qwen	flow	Y	
157	GEN-0 [52]	2025-11	Manip.	Other	10	n/d	n/d	N	
158	NORA-1.5 [213]	2025-11	Manip.	VLA	n/d	Qwen	hybrid	Y	960
159	SONIC [74]	2025-11	Hum.	VA	0.042	none	RL policy	Y	21,000
160	TWIST2 [214]	2025-11	Hum.	VA	n/d	none	n/d	?	
161	VIRAL [215]	2025-11	Hum.	VA	n/d	none	RL policy	?	
162	$\pi^*_{0.6}$ (RECAP) [88]	2025-11	Manip.	VLA	n/d	Gemma	hybrid	N	
163	$\pi_{0.6}$ [216]	2025-11	Manip.	VLA	n/d	Gemma	hybrid	N	
164	GR-Dexter [217]	2025-12	Manip.	VLA	4	Qwen	diffusion	?	
165	GR-RL [89]	2025-12	Manip.	VLA	5	Qwen	flow	?	
166	WholeBodyVLA [218]	2025-12	Hum.	VLA	n/d	Llama	latent	?	
167	Training-time RTC [219]	2025-12	Manip.	Other	n/d	Gemma	hybrid	N	
168	1X World Model policy [220]	2026-01	Hum.	Other	n/d	none	video WM	N	
169	Being-H0.5 [221]	2026-01	Manip.	VLA	n/d	n/d	flow	Y	
170	Cosmos Policy [32]	2026-01	Manip.	Other	2	none	video WM	Y	
171	G0Plus [222]	2026-01	Manip.	VLA	3	n/d	n/d	Y	
172	Helix 02 [223]	2026-01	Hum.	VLA	n/d	n/d	n/d	N	
173	InternVLA-A1 [224]	2026-01	Manip.	VLA	3.2	Qwen	flow	Y	
174	LingBot-VLA [225]	2026-01	Manip.	VLA	n/d	Qwen	flow	Y	
175	Rho-alpha ($\rho\alpha$) [226]	2026-01	Manip.	VLA	n/d	Phi	n/d	N	

#	Model	Date	Domain	Type	Params	LM	Head	W	GPU-h
176	Spirit v1.5 [227]	2026-01	Manip.	VLA	5	Qwen	n/d	Y	
177	UnifoLM-VLA-0 [228]	2026-01	Hum.	VLA	n/d	Qwen	n/d	Y	
178	VLingNav [82]	2026-01	Nav.	VLA	7	Qwen	regression	N	
179	ABot-N0 [81]	2026-02	Nav.	VLA	4	Qwen	flow	N	
180	DreamZero [33]	2026-02	Manip.	Other	14	none	video WM	Y	
181	EgoScale [229]	2026-02	Manip.	VLA	n/d	n/d	flow	?	
182	G0Tiny [230]	2026-02	Manip.	VLA	0.25	SmolLM	n/d	Y	
183	HoloBrain-0 [231]	2026-02	Manip.	VLA	1.1	Qwen	other	Y	
184	TIC-VLA [83]	2026-02	Nav.	VLA	n/d	Qwen	regression	Y	
185	Xiaomi-Robotics-0 [232]	2026-02	Manip.	VLA	4.7	Qwen	flow	Y	
186	GR00T N1.7 [51]	2026-03	Hum.	VLA	3	Qwen	flow	Y	30,720
187	SG-VLA [233]	2026-03	MoMa	VLA	1.3	Qwen	tokens	?	
188	Ψ_0 [234]	2026-03	Hum.	VLA	2.5	Qwen	hybrid	Y	15,360
189	Being-H0.7 [235]	2026-04	Manip.	Other	3	Qwen	n/d	?	
190	GEN-1 [53]	2026-04	Manip.	VLA	n/d	n/d	n/d	N	
191	GO-2 (Genie Operator-2) [236]	2026-04	Manip.	VLA	n/d	n/d	n/d	N	
192	$\pi_{0.7}$ [66]	2026-04	Manip.	VLA	5	Gemma	flow	N	
193	MolmoAct 2 [237]	2026-05	Manip.	VLA	5	n/d	flow	Y	5,760
194	Qwen-VLA [85]	2026-05	Nav.	VLA	n/d	Qwen	flow	N	
195	Uni-LaViRA [238]	2026-05	Nav.	Other	n/d	Gemini	other	P	
196	FutureNav [239]	2026-06	Nav.	VLA	8	Qwen	tokens	P	
197	G0.5 [240]	2026-06	MoMa	VLA	n/d	n/d	tokens	Y	
198	Galbot AstraBrain family [241]	2026-06	Hum.	Other	n/d	n/d	n/d	?	
199	OpenHLM [242]	2026-06	Hum.	VLA	n/d	Gemma	flow	Y	
200	Qwen-RobotNav [243]	2026-06	Nav.	VLA	8	Qwen	regression	N	2,816
201	SyVLA [90]	2026-06	Manip.	VLA	n/d	Qwen	flow	?	1,152
202	Gemini Robotics 2 [244]	2026-07	Hum.	VLA	n/d	Gemini	n/d	N	
203	Gemini Robotics On-Device 2 [245]	2026-07	Manip.	VLA	n/d	Gemma	n/d	N	
204	InternVLA-A1.5 [246]	2026-07	Manip.	VLA	n/d	Qwen	n/d	Y	
205	LingBot-VLA 2.0 [247]	2026-07	MoMa	VLA	6.4	Qwen	flow	Y	
206	TurboVLA [61]	2026-07	Manip.	VLA	0.2	n/d	n/d	?	
207	Xiaomi-Robotics-1 [248]	2026-07	MoMa	VLA	5.1	Qwen	flow	Y	
208	DYNA-2 [54]	2026-08	Manip.	Other	n/d	none	video WM	N	
209	GEN-1.5 [48]	2026-08	Manip.	VLA	n/d	n/d	n/d	N	
210	LightNav-0 [84]	2026-08	Nav.	VLA	4.8	Qwen	tokens	Y	1,120
211	PanoVLA [249]	2026-08	MoMa	VLA	n/d	Gemma	flow	?	
212	Skild S1 [250]	2026-08	Manip.	VA	n/d	n/d	n/d	N	
213	τ_0-VLA [251]	2026-08	MoMa	VLA	3	Qwen	flow	Y	
214	ω-0 (OMEGA-0) [252]	2026-08	Hum.	VLA	4	Qwen	diffusion	Y	
215	Spark-0 [253]	2026-10	Manip.	VLA	n/d	Qwen	flow	Y	

Appendix B. Estimator Inputs

Table A2. Device specifications used by the estimator (memory, memory bandwidth and dense BF16 tensor throughput). Values come from vendor specification pages or are derived from published SM counts and clocks; every value and its derivation is documented in the released dataset.

Device	Class	Memory (GB)	Bandwidth (GB/s)	BF16 (TFLOPS)
Jetson Orin Nano Super 8GB	Embedded	8	102	17
Jetson Orin NX 16GB	Embedded	16	102	19
Jetson AGX Orin 64GB	Embedded	64	205	43
Jetson AGX Thor (T5000)	Embedded	128	273	258.5
RTX 4060 Laptop	Laptop	8	256	29.1
RTX 4090 Laptop	Laptop	16	576	79.4
RTX 5080 Laptop	Laptop	16	896	70.3
RTX 3060 12GB	Desktop	12	360	25.5
RTX 3090	Desktop	24	936	71.2
RTX 4060 Ti 16GB	Desktop	16	288	44.1
RTX 4070 Ti SUPER	Desktop	16	672	88.2
RTX 4090	Desktop	24	1,008	165.2
RTX 5080	Desktop	16	960	112.6
RTX 5090	Desktop	32	1,792	209.5
RTX A6000	Workstation	48	768	154.8
RTX 6000 Ada	Workstation	48	960	364
RTX PRO 6000 Blackwell	Workstation	96	1,792	503.8
L40	Data centre	48	864	181.05
L40S	Data centre	48	864	362.05
A100 40GB SXM	Data centre	40	1,555	312
A100 80GB SXM	Data centre	80	2,039	312
H100 SXM	Data centre	80	3,350	989.5
H200 SXM	Data centre	141	4,800	989.5
B200	Data centre	180	8,000	2250

Table A3. Published latency measurements (anchors) used to calibrate and validate the estimator, with the calibrated estimate. Ratio = estimate/measurement. The video world-model policy (Cosmos Policy) has no estimate because its denoising cost is not modelled.

Model	Device	Prec.	Meas. (ms)	Est. (ms)	Ratio
OpenVLA [2]	RTX 4090	bf16	167	279	1.67
OpenVLA-OFT [95]	A100 80GB SXM	bf16	72.9	49	0.67
MiniVLA [146]	L40S	bf16	80	65	0.81
TinyVLA [62]	RTX A6000	bf16	14	71	5.05
SpatialVLA [63]	RTX 4090	bf16	50	233	4.67
VLA-0 [209]	RTX 5090	bf16	250	100	0.40
π_0 [3]	RTX 4090	bf16	73	90	1.24
π_0 [3]	Jetson AGX Orin 64GB	bf16	1280	349	0.27
π_0-FAST [153]	RTX 4090	bf16	750	667	0.89
$\pi_{0.5}$ [6]	RTX 4090	bf16	76	82	1.08
$\pi_{0.6}$ [216]	H100 SXM	bf16	63	62	0.98
RDT-1B [41]	RTX 4090	bf16	167	39	0.23
CogACT [142]	RTX A6000	bf16	181	145	0.80
FLOWER [198]	RTX 4090	bf16	52	26	0.51
Evo-1 [212]	RTX 4090	bf16	61	57	0.94
Octo [47]	RTX 4090	bf16	77	88	1.15
GR00T N1 [4]	L40	bf16	63.9	46	0.71
GR00T N1.5 [167]	H100 SXM	bf16	47.88	27	0.56
GR00T N1.6 [200]	RTX 5090	bf16	37	48	1.30

Model	Device	Prec.	Meas. (ms)	Est. (ms)	Ratio
GR00T N1.6 [200]	H100 SXM	bf16	38	27	0.70
GR00T N1.6 [200]	RTX 4090	bf16	44	57	1.29
GR00T N1.6 [200]	Jetson AGX Thor (T5000)	bf16	105	101	0.97
GR00T N1.7 [51]	H100 SXM	bf16	27.9	27	0.96
GR00T N1.7 [51]	RTX PRO 6000 Blackwell	bf16	27.9	33	1.18
GR00T N1.7 [51]	Jetson AGX Thor (T5000)	bf16	93.8	101	1.08
Cosmos Policy [32]	H100 SXM	bf16	610	–	–
Xiaomi-Robotics-0 [232]	RTX 4090	bf16	80	82	1.03
MolmoAct 2 [237]	H100 SXM	bf16	180	56	0.31
TurboVLA [61]	RTX 4090	bf16	31.2	5	0.15
Uni-NaVid [9]	A100 80GB SXM	bf16	200	178	0.89
NaVILA [8]	RTX 4090	bf16	594.6	490	0.82
NaVILA [8]	RTX 4090	int4	367.8	331	0.90
StreamVLN [188]	RTX 4090	bf16	270	369	1.37
TrackVLA [65]	RTX 4090	bf16	100	217	2.17
ABot-N0 [81]	Jetson Orin NX 16GB	bf16	500	1673	3.35
VLingNav [82]	RTX 4090	bf16	300	216	0.72
LightNav-0 [84]	Jetson AGX Thor (T5000)	bf16	268	621	2.32
LightNav-0 [84]	Jetson AGX Thor (T5000)	int8/fp8	178	399	2.24

References

1. Brohan, A.; Brown, N.; Carbajal, J.; Chebotar, Y.; Chen, X.; Choromanski, K.; Ding, T.; Driess, D.; Dubey, A.; Finn, C.; et al. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. *arXiv* **2023**, arXiv:2307.15818.
2. Kim, M.J.; Pertsch, K.; Karamcheti, S.; Xiao, T.; Balakrishna, A.; Nair, S.; Rafailov, R.; Foster, E.; Lam, G.; Sanketi, P.; et al. OpenVLA: An Open-Source Vision-Language-Action Model. *arXiv* **2024**, arXiv:2406.09246.
3. Black, K.; Brown, N.; Driess, D.; Esmail, A.; Equi, M.; Finn, C.; Fusai, N.; Groom, L.; Hausman, K.; Ichter, B.; et al. π_0: A Vision-Language-Action Flow Model for General Robot Control. *arXiv* **2024**, arXiv:2410.24164.
4. NVIDIA.; Bjorck, J.; Castañeda, F.; Cherniadev, N.; Da, X.; Ding, R.; Fan, L.; Fang, Y.; Fox, D.; Hu, F.; et al. GR00T N1: An Open Foundation Model for Generalist Humanoid Robots. *arXiv* **2025**, arXiv:2503.14734.
5. Figure AI. Helix: A Vision-Language-Action Model for Generalist Humanoid Control. Available online: https://www.figure.ai/news/helix (accessed on 6 October 2026).
6. Physical Intelligence.; Black, K.; Brown, N.; Darpinian, J.; Dhabalia, K.; Driess, D.; Esmail, A.; Equi, M.; Finn, C.; Fusai, N.; et al. $\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization. *arXiv* **2025**, arXiv:2504.16054.
7. Wu, Z.; Zhou, Y.; Xu, X.; Wang, Z.; Yan, H. MoManipVLA: Transferring Vision-language-action Models for General Mobile Manipulation. *arXiv* **2025**, arXiv:2503.13446.
8. Cheng, A.C.; Ji, Y.; Yang, Z.; Gongye, Z.; Zou, X.; Kautz, J.; Bıyık, E.; Yin, H.; Liu, S.; Wang, X. NaVILA: Legged Robot Vision-Language-Action Model for Navigation. *arXiv* **2024**, arXiv:2412.04453.
9. Zhang, J.; Wang, K.; Wang, S.; Li, M.; Liu, H.; Wei, S.; Wang, Z.; Zhang, Z.; Wang, H. Uni-NaVid: A Video-based Vision-Language-Action Model for Unifying Embodied Navigation Tasks. *arXiv* **2024**, arXiv:2412.06224.
10. Ma, Y.; Song, Z.; Zhuang, Y.; Hao, J.; King, I. A Survey on Vision-Language-Action Models for Embodied AI. *arXiv* **2024**, arXiv:2405.14093. https://doi.org/10.1109/TNNLS.2025.3650584.
11. Sapkota, R.; Cao, Y.; Roumeliotis, K.I.; Karkee, M. Vision-Language-Action (VLA) Models: Concepts, Progress, Applications and Challenges. *arXiv* **2025**, arXiv:2505.04769.
12. Zhong, Y.; Bai, F.; Cai, S.; Huang, X.; Chen, Z.; Zhang, X.; Wang, Y.; Guo, S.; Guan, T.; Lui, K.N.; et al. A Survey on Vision-Language-Action Models: An Action Tokenization Perspective. *arXiv* **2025**, arXiv:2507.01925.
13. Shao, R.; Li, W.; Zhang, L.; Zhang, R.; Liu, Z.; Chen, R.; Nie, L. Large VLM-based Vision-Language-Action Models for Robotic Manipulation: A Survey. *arXiv* **2025**, arXiv:2508.13073.
14. Kawaharazuka, K.; Oh, J.; Yamada, J.; Posner, I.; Zhu, Y. Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications. *arXiv* **2025**, arXiv:2510.07077. https://doi.org/10.1109/ACCESS.2025.3609980.
15. Guan, W.; Hu, Q.; Li, A.; Cheng, J. Efficient Vision-Language-Action Models for Embodied Manipulation: A Systematic Survey. *arXiv* **2025**, arXiv:2510.17111.
16. Yu, Z.; Wang, B.; Zeng, P.; Zhang, H.; Zhang, J.; Wang, Z.; Gao, L.; Song, J.; Sebe, N.; Shen, H.T. A Survey on Efficient Vision-Language-Action Models. *arXiv* **2025**, arXiv:2510.24795.

17. Grover, U.; Ranjan, R.; Mao, M.; Dong, T.T.; Praveen, S.; Wu, Z.; Chang, J.M.; Mohsenin, T.; Sheng, Y.; Polyzou, A.; et al. Embodied Foundation Models at the Edge: A Survey of Deployment Constraints and Mitigation Strategies. *arXiv* **2026**, arXiv:2603.16952.
18. Wang, Z.; Wang, B.; Zhang, H.; Du, T.; Chen, T.; Sun, G.; He, Y.; Shen, Z.; Ye, W.; Li, A. Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines. *arXiv* **2026**, arXiv:2604.23001.
19. Firoozi, R.; Tucker, J.; Tian, S.; Majumdar, A.; Sun, J.; Liu, W.; Zhu, Y.; Song, S.; Kapoor, A.; Hausman, K.; et al. Foundation Models in Robotics: Applications, Challenges, and the Future. *arXiv* **2023**, arXiv:2312.07843.
20. Hu, Y.; Xie, Q.; Jain, V.; Francis, J.; Patrikar, J.; Keetha, N.; Kim, S.; Xie, Y.; Zhang, T.; Fang, H.S.; et al. Toward General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis. *arXiv* **2023**, arXiv:2312.08782.
21. Kawaharazuka, K.; Matsushima, T.; Gambardella, A.; Guo, J.; Paxton, C.; Zeng, A. Real-World Robot Applications of Foundation Models: A Review. *arXiv* **2024**, arXiv:2402.05741. https://doi.org/10.1080/01691864.2024.2408593.
22. Adilkhanov, A.; Yelenov, A.; Seitzhanov, A.; Mazhitov, A.; Abdikarimov, A.; Sandykbayeva, D.; Kenzhebek, D.; Mukashev, D.; Umurbekov, I.; Chumakov, J.; et al. Survey on Vision-Language-Action Models. *arXiv* **2025**, arXiv:2502.06851.
23. Din, M.U.; Akram, W.; Saoud, L.S.; Rosell, J.; Hussain, I. Vision Language Action Models in Robotic Manipulation: A Systematic Review. *arXiv* **2025**, arXiv:2507.10672.
24. Li, H.; Chen, Y.; Cui, W.; Liu, W.; Liu, K.; Zhou, M.; Zhang, Z.; Zhao, D. Survey of Vision-Language-Action Models for Embodied Manipulation. *arXiv* **2025**, arXiv:2508.15201.
25. Zhang, D.; Sun, J.; Hu, C.; Wu, X.; Yuan, Z.; Zhou, R.; Shen, F.; Zhou, Q. Pure Vision Language Action (VLA) Models: A Comprehensive Survey. *arXiv* **2025**, arXiv:2509.19012.
26. Shen, Q.; Zhang, S.; Liao, Y.; Li, Q.; Tan, Z.; Wang, S.; Yan, S.; Wang, X. World Action Models: A Survey. *arXiv* **2026**, arXiv:2606.20781.
27. Lu, Z.; Zhai, H.; Wang, G.; Zeng, H.; Yang, J.; Liu, J.; Cheng, L.; Zhang, Y.; Qiu, Y.; Cheng, Z.; et al. World-Action Models for Robot Learning and Control: A Survey. *arXiv* **2026**, arXiv:2609.16074.
28. Lisondra, M.; Benhabib, B.; Nejat, G. Embodied AI with Foundation Models for Mobile Service Robots: A Systematic Review. *arXiv* **2025**, arXiv:2505.20503. https://doi.org/10.3390/robotics15030055.
29. Sa, I.; Park, C.; Lee, H.M.; Noh, D.; Ahn, H.S. Vision Language Action (VLA) Models for Unmanned Aerial Robotics and Bimanual Manipulation: A Review. *arXiv* **2026**, arXiv:2607.06706. https://doi.org/10.3390/drones10060412.
30. Zhao, T.Z.; Kumar, V.; Levine, S.; Finn, C. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. *arXiv* **2023**, arXiv:2304.13705.
31. Chi, C.; Xu, Z.; Feng, S.; Cousineau, E.; Du, Y.; Burchfiel, B.; Tedrake, R.; Song, S. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. *arXiv* **2023**, arXiv:2303.04137.
32. Kim, M.J.; Gao, Y.; Lin, T.Y.; Lin, Y.C.; Ge, Y.; Lam, G.; Liang, P.; Song, S.; Liu, M.Y.; Finn, C.; et al. Cosmos Policy: Fine-Tuning Video Models for Visuomotor Control and Planning. *arXiv* **2026**, arXiv:2601.16163.
33. Ye, S.; Ge, Y.; Zheng, K.; Gao, S.; Yu, S.; Kurian, G.; Indupuru, S.; Tan, Y.L.; Zhu, C.; Xiang, J.; et al. World Action Models are Zero-shot Policies. *arXiv* **2026**, arXiv:2602.15922.
34. Gemini Robotics Team.; Abeyruwan, S.; Ainslie, J.; Alayrac, J.B.; Arenas, M.G.; Armstrong, T.; Balakrishna, A.; Baruch, R.; Bauza, M.; Blokzijl, M.; et al. Gemini Robotics: Bringing AI into the Physical World. *arXiv* **2025**, arXiv:2503.20020.
35. Black, K.; Galliker, M.Y.; Levine, S. Real-Time Execution of Action Chunking Flow Policies. *arXiv* **2025**, arXiv:2506.07339.
36. Ahn, M.; Dwibedi, D.; Finn, C.; Arenas, M.G.; Gopalakrishnan, K.; Hausman, K.; Ichter, B.; Irpan, A.; Joshi, N.; Julian, R.; et al. AutoRT: Embodied Foundation Models for Large Scale Orchestration of Robotic Agents. *arXiv* **2024**, arXiv:2401.12963.
37. Brohan, A.; Brown, N.; Carbajal, J.; Chebotar, Y.; Dabis, J.; Finn, C.; Gopalakrishnan, K.; Hausman, K.; Herzog, A.; Hsu, J.; et al. RT-1: Robotics Transformer for Real-World Control at Scale. *arXiv* **2022**, arXiv:2212.06817.
38. Lipman, Y.; Chen, R.T.Q.; Ben-Hamu, H.; Nickel, M.; Le, M. Flow Matching for Generative Modeling. *arXiv* **2022**, arXiv:2210.02747.
39. Ho, J.; Jain, A.; Abbeel, P. Denoising Diffusion Probabilistic Models. *arXiv* **2020**, arXiv:2006.11239.
40. Beyer, L.; Steiner, A.; Pinto, A.S.; Kolesnikov, A.; Wang, X.; Salz, D.; Neumann, M.; Alabdulmohsin, I.; Tschannen, M.; Bugliarello, E.; et al. PaliGemma: A versatile 3B VLM for transfer. *arXiv* **2024**, arXiv:2407.07726.
41. Liu, S.; Wu, L.; Li, B.; Tan, H.; Chen, H.; Wang, Z.; Xu, K.; Su, H.; Zhu, J. RDT-1B: a Diffusion Foundation Model for Bimanual Manipulation. *arXiv* **2024**, arXiv:2410.07864.
42. Karamcheti, S.; Nair, S.; Balakrishna, A.; Liang, P.; Kollar, T.; Sadigh, D. Prismatic VLMs: Investigating the Design Space of Visually-Conditioned Language Models. *arXiv* **2024**, arXiv:2402.07865.
43. Bai, S.; Chen, K.; Liu, X.; Wang, J.; Ge, W.; Song, S.; Dang, K.; Wang, P.; Wang, S.; Tang, J.; et al. Qwen2.5-VL Technical Report. *arXiv* **2025**, arXiv:2502.13923.
44. Open X-Embodiment Collaboration.; O'Neill, A.; Rehman, A.; Gupta, A.; Maddukuri, A.; Gupta, A.; Padalkar, A.; Lee, A.; Pooley, A.; Gupta, A.; et al. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. *arXiv* **2023**, arXiv:2310.08864.
45. Belkhale, S.; Ding, T.; Xiao, T.; Sermanet, P.; Vuong, Q.; Tompson, J.; Chebotar, Y.; Dwibedi, D.; Sadigh, D. RT-H: Action Hierarchies Using Language. *arXiv* **2024**, arXiv:2403.01823.

46. Wen, J.; Zhu, M.; Zhu, Y.; Tang, Z.; Li, J.; Zhou, Z.; Li, C.; Liu, X.; Peng, Y.; Shen, C.; et al. Diffusion-VLA: Generalizable and Interpretable Robot Foundation Model via Self-Generated Reasoning. *arXiv* **2024**, arXiv:2412.03293.
47. Octo Model Team.; Ghosh, D.; Walke, H.; Pertsch, K.; Black, K.; Mees, O.; Dasari, S.; Hejna, J.; Kreiman, T.; Xu, C.; et al. Octo: An Open-Source Generalist Robot Policy. *arXiv* **2024**, arXiv:2405.12213.
48. Generalist AI. GEN-1.5: Embodied Foundation Models are One-Shot Learners. Available online: https://generalistai.com/blog/gen-1.5 (accessed on 6 October 2026).
49. Makoviychuk, V.; Wawrzyniak, L.; Guo, Y.; Lu, M.; Storey, K.; Macklin, M.; Hoeller, D.; Rudin, N.; Allshire, A.; Handa, A.; et al. Isaac Gym: High Performance GPU-Based Physics Simulation For Robot Learning. *arXiv* **2021**, arXiv:2108.10470.
50. He, T.; Xiao, W.; Lin, T.; Luo, Z.; Xu, Z.; Jiang, Z.; Kautz, J.; Liu, C.; Shi, G.; Wang, X.; et al. HOVER: Versatile Neural Whole-Body Controller for Humanoid Robots. *arXiv* **2024**, arXiv:2410.21229.
51. NVIDIA. GR00T-N1.7-3B (Hugging Face model card). Available online: https://huggingface.co/nvidia/GR00T-N1.7-3B (accessed on 6 October 2026).
52. Generalist AI. GEN-0. Available online: https://generalistai.com/blog/nov-04-2025-GEN-0 (accessed on 6 October 2026).
53. Generalist AI. GEN-1: Scaling Embodied Foundation Models to Mastery. Available online: https://generalistai.com/blog/apr-02-2026-GEN-1 (accessed on 6 October 2026).
54. Dyna Robotics. Dyna-2: A 1-Million-Hour Scaling Law for World-Action Models. Available online: https://www.dyna.co/dyna-2 (accessed on 6 October 2026).
55. Assran, M.; Bardes, A.; Fan, D.; Garrido, Q.; Howes, R.; Komeili, M.; Muckley, M.; Rizvi, A.; Roberts, C.; Sinha, K.; et al. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. *arXiv* **2025**, arXiv:2506.09985.
56. Lin, F.; Hu, Y.; Sheng, P.; Wen, C.; You, J.; Gao, Y. Data Scaling Laws in Imitation Learning for Robotic Manipulation. *arXiv* **2024**, arXiv:2410.18647.
57. Khazatsky, A.; Pertsch, K.; Nair, S.; Balakrishna, A.; Dasari, S.; Karamcheti, S.; Nasiriany, S.; Srirama, M.K.; Chen, L.Y.; Ellis, K.; et al. DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset. *arXiv* **2024**, arXiv:2403.12945.
58. Lin, J.; Tang, J.; Tang, H.; Yang, S.; Chen, W.M.; Wang, W.C.; Xiao, G.; Dang, X.; Gan, C.; Han, S. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. *arXiv* **2023**, arXiv:2306.00978.
59. Williams, S.; Waterman, A.; Patterson, D. Roofline: An Insightful Visual Performance Model for Multicore Architectures. *Commun. ACM* **2009**, *52*, 65–76. https://doi.org/10.1145/1498765.1498785.
60. Pope, R.; Douglas, S.; Chowdhery, A.; Devlin, J.; Bradbury, J.; Levskaya, A.; Heek, J.; Xiao, K.; Agrawal, S.; Dean, J. Efficiently Scaling Transformer Inference. *arXiv* **2022**, arXiv:2211.05102.
61. Xie, H.; Yao, C.; Wu, X.; Zhu, Y.; Liang, D.; Bai, X.; Ding, H. TurboVLA: Real-Time Vision-Language-Action Model at 32 Hz on an RTX 4090 with <1 GB VRAM. *arXiv* **2026**, arXiv:2607.27205.
62. Wen, J.; Zhu, Y.; Li, J.; Zhu, M.; Wu, K.; Xu, Z.; Liu, N.; Cheng, R.; Shen, C.; Peng, Y.; et al. TinyVLA: Towards Fast, Data-Efficient Vision-Language-Action Models for Robotic Manipulation. *arXiv* **2024**, arXiv:2409.12514.
63. Qu, D.; Song, H.; Chen, Q.; Yao, Y.; Ye, X.; Ding, Y.; Wang, Z.; Gu, J.; Zhao, B.; Wang, D.; et al. SpatialVLA: Exploring Spatial Representations for Visual-Language-Action Model. *arXiv* **2025**, arXiv:2501.15830.
64. Wang, L.; Chen, X.; Zhao, J.; He, K. Scaling Proprioceptive-Visual Learning with Heterogeneous Pre-trained Transformers. *arXiv* **2024**, arXiv:2409.20537.
65. Wang, S.; Zhang, J.; Li, M.; Liu, J.; Li, A.; Wu, K.; Zhong, F.; Yu, J.; Zhang, Z.; Wang, H. TrackVLA: Embodied Visual Tracking in the Wild. *arXiv* **2025**, arXiv:2505.23189.
66. Physical Intelligence.; Ai, B.; Amin, A.; Aniceto, R.; Balakrishna, A.; Balke, G.; Black, K.; Bokinsky, G.; Cao, S.; Charbonnier, T.; et al. $\pi_{0.7}$: a Steerable Generalist Robotic Foundation Model with Emergent Capabilities. *arXiv* **2026**, arXiv:2604.15483.
67. Rajbhandari, S.; Rasley, J.; Ruwase, O.; He, Y. ZeRO: Memory Optimizations Toward Training Trillion Parameter Models. *arXiv* **2019**, arXiv:1910.02054.
68. Zhao, Y.; Gu, A.; Varma, R.; Luo, L.; Huang, C.C.; Xu, M.; Wright, L.; Shojanazeri, H.; Ott, M.; Shleifer, S.; et al. PyTorch FSDP: Experiences on Scaling Fully Sharded Data Parallel. *arXiv* **2023**, arXiv:2304.11277.
69. Hu, E.J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; Chen, W. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv* **2021**, arXiv:2106.09685.
70. Dettmers, T.; Pagnoni, A.; Holtzman, A.; Zettlemoyer, L. QLoRA: Efficient Finetuning of Quantized LLMs. *arXiv* **2023**, arXiv:2305.14314.
71. AgiBot World Contributors.; Bu, Q.; Cai, J.; Chen, L.; Cui, X.; Ding, Y.; Feng, S.; Gao, S.; He, X.; Hu, X.; et al. AgiBot World Colosseo: A Large-scale Manipulation Platform for Scalable and Intelligent Embodied Systems. *arXiv* **2025**, arXiv:2503.06669.
72. Kaplan, J.; McCandlish, S.; Henighan, T.; Brown, T.B.; Chess, B.; Child, R.; Gray, S.; Radford, A.; Wu, J.; Amodei, D. Scaling Laws for Neural Language Models. *arXiv* **2020**, arXiv:2001.08361.
73. Liao, Q.; Truong, T.E.; Huang, X.; Gao, Y.; Tevet, G.; Sreenath, K.; Liu, C.K. BeyondMimic: From Motion Tracking to Versatile Humanoid Control via Guided Diffusion. *arXiv* **2025**, arXiv:2508.08241.

74. Luo, Z.; Yuan, Y.; Wang, T.; Li, C.; Castañeda, F.; Chen, S.; Cao, Z.A.; Li, J.; Minor, D.; Ben, Q.; et al. SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control. *arXiv* **2025**, arXiv:2511.07820. https://doi.org/10.1126/scirobotics.aed4592.
75. Xue, H.; Huang, X.; Niu, D.; Liao, Q.; Kragerud, T.; Gravdahl, J.T.; Peng, X.B.; Shi, G.; Darrell, T.; Sreenath, K.; et al. LeVERB: Humanoid Whole-Body Control with Latent Vision-Language Instruction. *arXiv* **2025**, arXiv:2506.13751.
76. Liu, M.; Chen, Z.; Cheng, X.; Ji, Y.; Qiu, R.Z.; Yang, R.; Wang, X. Visual Whole-Body Control for Legged Loco-Manipulation. *arXiv* **2024**, arXiv:2403.16967.
77. Pan, G.; Ben, Q.; Yuan, Z.; Jiang, G.; Ji, Y.; Li, S.; Pang, J.; Liu, H.; Xu, H. RoboDuet: Learning a Cooperative Policy for Whole-body Legged Loco-Manipulation. *arXiv* **2024**, arXiv:2403.17367.
78. Liu, W.; Zhao, H.; Li, C.; Deng, Y.; Biswas, J.; Pouya, S.; Chang, Y. COMPASS: Cross-embodiment Mobility Policy via Residual RL and Skill Synthesis. *arXiv* **2025**, arXiv:2502.16372.
79. Eftekhar, A.; Hendrix, R.; Weihs, L.; Duan, J.; Caglar, E.; Salvador, J.; Herrasti, A.; Han, W.; VanderBil, E.; Kembhavi, A.; et al. The One RING: a Robotic Indoor Navigation Generalist. *arXiv* **2024**, arXiv:2412.14401.
80. Qi, Z.; Zhang, Z.; Yu, Y.; Wang, J.; Zhao, H. VLN-R1: Vision-Language Navigation via Reinforcement Fine-Tuning. *arXiv* **2025**, arXiv:2506.17221.
81. Chu, Z.; Xie, S.; Wu, X.; Shen, Y.; Luo, M.; Wang, Z.; Liu, F.; Leng, X.; Hu, J.; Yin, M.; et al. ABot-N0: Technical Report on the VLA Foundation Model for Versatile Embodied Navigation. *arXiv* **2026**, arXiv:2602.11598.
82. Wang, S.; Luo, Y.; Chen, X.; Luo, A.; Li, D.; Liu, C.; Chen, S.; Zhang, Y.; Yu, J. VLingNav: Embodied Navigation with Adaptive Reasoning and Visual-Assisted Linguistic Memory. *arXiv* **2026**, arXiv:2601.08665.
83. Huang, Z.; Zhang, Y.; Liu, J.; Song, R.; Tang, C.; Ma, J. TIC-VLA: A Think-in-Control Vision-Language-Action Model for Robot Navigation in Dynamic Environments. *arXiv* **2026**, arXiv:2602.02459.
84. Wang, S.; Luo, A.; Huang, F.; Xu, J.; Wang, X.; Wang, Y.; Ma, Q.; Yang, F.; Mei, R.; Wei, J.; et al. LightNav-0: Eliciting VLM Spatial Intelligence for Generalist Embodied Navigation. *arXiv* **2026**, arXiv:2608.30935.
85. Wang, Q.; Li, M.; Guan, J.; Ye, J.; Xie, S.; Liu, Y.; Chen, J.; Liang, Z.; Zhang, J.; Hu, X.; et al. Qwen-VLA: Unifying Vision-Language-Action Modeling across Tasks, Environments, and Robot Embodiments. *arXiv* **2026**, arXiv:2605.30280.
86. Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; Klimov, O. Proximal Policy Optimization Algorithms. *arXiv* **2017**, arXiv:1707.06347.
87. Shao, Z.; Wang, P.; Zhu, Q.; Xu, R.; Song, J.; Bi, X.; Zhang, H.; Zhang, M.; Li, Y.K.; Wu, Y.; et al. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *arXiv* **2024**, arXiv:2402.03300.
88. Physical Intelligence.; Amin, A.; Aniceto, R.; Balakrishna, A.; Black, K.; Conley, K.; Connors, G.; Darpinian, J.; Dhabalia, K.; DiCarlo, J.; et al. $\pi^*_{0.6}$: a VLA That Learns From Experience. *arXiv* **2025**, arXiv:2511.14759.
89. Li, Y.; Ma, X.; Xu, J.; Cui, Y.; Cui, Z.; Han, Z.; Huang, L.; Kong, T.; Liu, Y.; Niu, H.; et al. GR-RL: Going Dexterous and Precise for Long-Horizon Robotic Manipulation. *arXiv* **2025**, arXiv:2512.01801.
90. Wang, L.; Wang, Z.; Li, X.; Fan, L.; Lu, C.; Ye, N. Scaling by Diversified Experience for Vision-Language-Action Models. *arXiv* **2026**, arXiv:2606.09009.
91. Zhao, H.; Song, W.; Wang, D.; Tong, X.; Ding, P.; Cheng, X.; Ge, Z. MoRE: Unlocking Scalability in Reinforcement Learning for Quadruped Vision-Language-Action Models. *arXiv* **2025**, arXiv:2503.08007.
92. Shah, D.; Sridhar, A.; Bhorkar, A.; Hirose, N.; Levine, S. GNM: A General Navigation Model to Drive Any Robot. *arXiv* **2022**, arXiv:2210.03370.
93. Shah, D.; Sridhar, A.; Dashora, N.; Stachowicz, K.; Black, K.; Hirose, N.; Levine, S. ViNT: A Foundation Model for Visual Navigation. *arXiv* **2023**, arXiv:2306.14846.
94. Sridhar, A.; Shah, D.; Glossop, C.; Levine, S. NoMaD: Goal Masked Diffusion Policies for Navigation and Exploration. *arXiv* **2023**, arXiv:2310.07896.
95. Kim, M.J.; Finn, C.; Liang, P. Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success. *arXiv* **2025**, arXiv:2502.19645.
96. Mitchell, M.; Wu, S.; Zaldivar, A.; Barnes, P.; Vasserman, L.; Hutchinson, B.; Spitzer, E.; Raji, I.D.; Gebru, T. Model Cards for Model Reporting. *arXiv* **2018**, arXiv:1810.03993. https://doi.org/10.1145/3287560.3287596.
97. Gebru, T.; Morgenstern, J.; Vecchione, B.; Vaughan, J.W.; Wallach, H.; Daumé, H.; Crawford, K. Datasheets for Datasets. *arXiv* **2018**, arXiv:1803.09010.
98. Pineau, J.; Vincent-Lamarre, P.; Sinha, K.; Larivière, V.; Beygelzimer, A.; d'Alché Buc, F.; Fox, E.; Larochelle, H. Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program). *arXiv* **2020**, arXiv:2003.12206.
99. Strubell, E.; Ganesh, A.; McCallum, A. Energy and Policy Considerations for Deep Learning in NLP. *arXiv* **2019**, arXiv:1906.02243.
100. Schwartz, R.; Dodge, J.; Smith, N.A.; Etzioni, O. Green AI. *arXiv* **2019**, arXiv:1907.10597.
101. Henderson, P.; Hu, J.; Romoff, J.; Brunskill, E.; Jurafsky, D.; Pineau, J. Towards the Systematic Reporting of the Energy and Carbon Footprints of Machine Learning. *arXiv* **2020**, arXiv:2002.05651.

102. Reddi, V.J.; Cheng, C.; Kanter, D.; Mattson, P.; Schmuelling, G.; Wu, C.J.; Anderson, B.; Breughe, M.; Charlebois, M.; Chou, W.; et al. MLPerf Inference Benchmark. *arXiv* **2019**, arXiv:1911.02549.
103. Li, X.; Liu, M.; Zhang, H.; Yu, C.; Xu, J.; Wu, H.; Cheang, C.; Jing, Y.; Zhang, W.; Liu, H.; et al. Vision-Language Foundation Models as Effective Robot Imitators. *arXiv* **2023**, arXiv:2311.01378.
104. Wu, H.; Jing, Y.; Cheang, C.; Chen, G.; Xu, J.; Li, X.; Liu, M.; Li, H.; Kong, T. Unleashing Large-Scale Video Generative Pre-training for Visual Robot Manipulation. *arXiv* **2023**, arXiv:2312.13139.
105. Yang, R.; Kim, Y.; Hendrix, R.; Kembhavi, A.; Wang, X.; Ehsani, K. Harmonic Mobile Manipulation. *arXiv* **2023**, arXiv:2312.06639.
106. Ding, P.; Zhao, H.; Zhang, W.; Song, W.; Zhang, M.; Huang, S.; Yang, N.; Wang, D. QUAR-VLA: Vision-Language-Action Model for Quadruped Robots. *arXiv* **2023**, arXiv:2312.14457.
107. Leal, I.; Choromanski, K.; Jain, D.; Dubey, A.; Varley, J.; Ryoo, M.; Lu, Y.; Liu, F.; Sindhwani, V.; Vuong, Q.; et al. SARA-RT: Scaling up Robotics Transformers with Self-Adaptive Robust Attention. *arXiv* **2023**, arXiv:2312.01990.
108. Fu, Z.; Zhao, T.Z.; Finn, C. Mobile ALOHA: Learning Bimanual Mobile Manipulation with Low-Cost Whole-Body Teleoperation. *arXiv* **2024**, arXiv:2401.02117.
109. Radosavovic, I.; Zhang, B.; Shi, B.; Rajasegaran, J.; Kamat, S.; Darrell, T.; Sreenath, K.; Malik, J. Humanoid Locomotion as Next Token Prediction. *arXiv* **2024**, arXiv:2402.19469.
110. Zhang, J.; Wang, K.; Xu, R.; Zhou, G.; Hong, Y.; Fang, X.; Wu, Q.; Zhang, Z.; Wang, H. NaVid: Video-based VLM Plans the Next Step for Vision-and-Language Navigation. *arXiv* **2024**, arXiv:2402.15852.
111. Chi, C.; Xu, Z.; Pan, C.; Cousineau, E.; Burchfiel, B.; Feng, S.; Tedrake, R.; Song, S. Universal Manipulation Interface: In-The-Wild Robot Teaching Without In-The-Wild Robots. *arXiv* **2024**, arXiv:2402.10329.
112. Zhen, H.; Qiu, X.; Chen, P.; Yang, J.; Yan, X.; Du, Y.; Hong, Y.; Gan, C. 3D-VLA: A 3D Vision-Language-Action Generative World Model. *arXiv* **2024**, arXiv:2403.09631.
113. Ze, Y.; Zhang, G.; Zhang, K.; Hu, C.; Wang, M.; Xu, H. 3D Diffusion Policy: Generalizable Visuomotor Policy Learning via Simple 3D Representations. *arXiv* **2024**, arXiv:2403.03954.
114. He, Z.; Lei, K.; Ze, Y.; Sreenath, K.; Li, Z.; Xu, H. Learning Visual Quadrupedal Loco-Manipulation from Demonstrations. *arXiv* **2024**, arXiv:2403.20328.
115. Covariant. Introducing RFM-1: Giving Robots Human-like Reasoning Capabilities. Available online: https://covariant.ai/insights/introducing-rfm-1-giving-robots-human-like-reasoning-capabilities/ (accessed on 6 October 2026).
116. Lee, S.; Wang, Y.; Etukuru, H.; Kim, H.J.; Shafiullah, N.M.M.; Pinto, L. Behavior Generation with Latent Actions. *arXiv* **2024**, arXiv:2403.03181.
117. Wang, C.; Fang, H.; Fang, H.S.; Lu, C. RISE: 3D Perception Makes Real-World Robot Imitation Simple and Effective. *arXiv* **2024**, arXiv:2404.12281.
118. Tesla. AI & Robotics. Available online: https://www.tesla.com/AI (accessed on 6 October 2026).
119. Haldar, S.; Peng, Z.; Pinto, L. BAKU: An Efficient Transformer for Multi-Task Policy Learning. *arXiv* **2024**, arXiv:2406.07539.
120. Fu, Z.; Zhao, Q.; Wu, Q.; Wetzstein, G.; Finn, C. HumanPlus: Humanoid Shadowing and Imitation from Humans. *arXiv* **2024**, arXiv:2406.10454.
121. Li, X.; Mata, C.; Park, J.; Kahatapitiya, K.; Jang, Y.S.; Shang, J.; Ranasinghe, K.; Burgert, R.; Cai, M.; Lee, Y.J.; et al. LLaRA: Supercharging Robot Learning Data for Vision-Language Policy. *arXiv* **2024**, arXiv:2406.20095.
122. He, T.; Luo, Z.; He, X.; Xiao, W.; Zhang, C.; Zhang, W.; Kitani, K.; Liu, C.; Shi, G. OmniH2O: Universal and Dexterous Human-to-Humanoid Whole-Body Teleoperation and Learning. *arXiv* **2024**, arXiv:2406.08858.
123. Liu, J.; Liu, M.; Wang, Z.; An, P.; Li, X.; Zhou, K.; Yang, S.; Zhang, R.; Guo, Y.; Zhang, S. RoboMamba: Efficient Vision-Language-Action Model for Robotic Reasoning and Manipulation. *arXiv* **2024**, arXiv:2406.04339.
124. Zawalski, M.; Chen, W.; Pertsch, K.; Mees, O.; Finn, C.; Levine, S. Robotic Control via Embodied Chain-of-Thought Reasoning. *arXiv* **2024**, arXiv:2407.08693.
125. Chiang, H.T.L.; Xu, Z.; Fu, Z.; Jacob, M.G.; Zhang, T.; Lee, T.W.E.; Yu, W.; Schenck, C.; Rendleman, D.; Shah, D.; et al. Mobility VLA: Multimodal Instruction Navigation with Long-Context VLMs and Topological Graphs. *arXiv* **2024**, arXiv:2407.07775.
126. Ha, H.; Gao, Y.; Fu, Z.; Tan, J.; Song, S. UMI on Legs: Making Manipulation Policies Mobile with Manipulation-Centric Whole-body Controllers. *arXiv* **2024**, arXiv:2407.10353.
127. Doshi, R.; Walke, H.; Mees, O.; Dasari, S.; Levine, S. Scaling Cross-Embodied Learning: One Policy for Manipulation, Navigation, Locomotion and Aviation. *arXiv* **2024**, arXiv:2408.11812.
128. Wu, Q.; Fu, Z.; Cheng, X.; Wang, X.; Finn, C. Helpful DoggyBot: Open-World Object Fetching using Legged Robots and Vision-Language Models. *arXiv* **2024**, arXiv:2410.00231.
129. Lee, A.; Chuang, I.; Chen, L.Y.; Soltani, I. InterACT: Inter-dependency Aware Action Chunking with Hierarchical Attention Transformers for Bimanual Manipulation. *arXiv* **2024**, arXiv:2409.07914.
130. Etukuru, H.; Naka, N.; Hu, Z.; Lee, S.; Mehu, J.; Edsinger, A.; Paxton, C.; Chintala, S.; Pinto, L.; Shafiullah, N.M.M. Robot Utility Models: General Policies for Zero-Shot Deployment in New Environments. *arXiv* **2024**, arXiv:2409.05865.

131. Zhu, M.; Zhu, Y.; Li, J.; Wen, J.; Xu, Z.; Liu, N.; Cheng, R.; Shen, C.; Peng, Y.; Feng, F.; et al. Scaling Diffusion Policy in Transformer to 1 Billion Parameters for Robotic Manipulation. *arXiv* **2024**, arXiv:2409.14411.
132. Zhao, T.Z.; Tompson, J.; Driess, D.; Florence, P.; Ghasemipour, K.; Finn, C.; Wahid, A. ALOHA Unleashed: A Simple Recipe for Robot Dexterity. *arXiv* **2024**, arXiv:2410.13126.
133. Radosavovic, I.; Kamat, S.; Darrell, T.; Malik, J. Learning Humanoid Locomotion over Challenging Terrain. *arXiv* **2024**, arXiv:2410.03654.
134. Hou, Z.; Zhang, T.; Xiong, Y.; Duan, H.; Pu, H.; Tong, R.; Zhao, C.; Zhu, X.; Qiao, Y.; Dai, J.; et al. Dita: Scaling Diffusion Transformer for Generalist Vision-Language-Action Policy. *arXiv* **2025**, arXiv:2503.19757.
135. Kareer, S.; Patel, D.; Punamiya, R.; Mathur, P.; Cheng, S.; Wang, C.; Hoffman, J.; Xu, D. EgoMimic: Scaling Imitation Learning via Egocentric Video. *arXiv* **2024**, arXiv:2410.24221.
136. Cheang, C.L.; Chen, G.; Jing, Y.; Kong, T.; Li, H.; Li, Y.; Liu, Y.; Wu, H.; Xu, J.; Yang, Y.; et al. GR-2: A Generative Video-Language-Action Model with Web-Scale Knowledge for Robot Manipulation. *arXiv* **2024**, arXiv:2410.06158.
137. Ze, Y.; Chen, Z.; Wang, W.; Chen, T.; He, X.; Yuan, Y.; Peng, X.B.; Wu, J. Generalizable Humanoid Manipulation with 3D Diffusion Policies. *arXiv* **2024**, arXiv:2410.10803.
138. Ye, S.; Jang, J.; Jeon, B.; Joo, S.; Yang, J.; Peng, B.; Mandlekar, A.; Tan, R.; Chao, Y.W.; Lin, B.Y.; et al. Latent Action Pretraining from Videos. *arXiv* **2024**, arXiv:2410.11758.
139. Hirose, N.; Glossop, C.; Sridhar, A.; Shah, D.; Mees, O.; Levine, S. LeLaN: Learning A Language-Conditioned Navigation Policy from In-the-Wild Videos. *arXiv* **2024**, arXiv:2410.03603.
140. Liu, W.; Zhao, H.; Li, C.; Biswas, J.; Okal, B.; Goyal, P.; Chang, Y.; Pouya, S. X-MOBILITY: End-To-End Generalizable Navigation via World Modeling. *arXiv* **2024**, arXiv:2410.17491.
141. Liu, X.; Li, J.; Jiang, Y.; Sujay, N.; Yang, Z.; Zhang, J.; Abanes, J.; Zhang, J.; Feng, C. CityWalker: Learning Embodied Urban Navigation from Web-Scale Videos. *arXiv* **2024**, arXiv:2411.17820.
142. Li, Q.; Liang, Y.; Wang, Z.; Luo, L.; Chen, X.; Liao, M.; Wei, F.; Deng, Y.; Xu, S.; Zhang, Y.; et al. CogACT: A Foundational Vision-Language-Action Model for Synergizing Cognition and Action in Robotic Manipulation. *arXiv* **2024**, arXiv:2411.19650.
143. Wu, J.; Chong, W.; Holmberg, R.; Prasad, A.; Gao, Y.; Khatib, O.; Song, S.; Rusinkiewicz, S.; Bohg, J. TidyBot++: An Open-Source Holonomic Mobile Manipulator for Robot Learning. *arXiv* **2024**, arXiv:2412.10447.
144. Gong, Z.; Ding, P.; Lyu, S.; Huang, S.; Sun, M.; Zhao, W.; Fan, Z.; Wang, D. CARP: Visuomotor Policy Learning via Coarse-to-Fine Autoregressive Prediction. *arXiv* **2024**, arXiv:2412.06782.
145. Ji, M.; Peng, X.; Liu, F.; Li, J.; Yang, G.; Cheng, X.; Wang, X. ExBody2: Advanced Expressive Humanoid Whole-Body Control. *arXiv* **2024**, arXiv:2412.13196.
146. Stanford ILIAD. MiniVLA (Stanford AI Lab blog post). Available online: https://ai.stanford.edu/blog/minivla/ (accessed on 6 October 2026).
147. Chen, Y.; Ge, Y.; Tang, W.; Li, Y.; Ge, Y.; Ding, M.; Shan, Y.; Liu, X. Moto: Latent Motion Token as the Bridging Language for Learning Robot Manipulation from Videos. *arXiv* **2024**, arXiv:2412.04445.
148. Bar, A.; Zhou, G.; Tran, D.; Darrell, T.; LeCun, Y. Navigation World Models. *arXiv* **2024**, arXiv:2412.03572.
149. Tong, X.; Ding, P.; Fan, Y.; Wang, D.; Zhang, W.; Cui, C.; Sun, M.; Zhao, H.; Zhang, H.; Dang, Y.; et al. QUART-Online: Latency-Free Large Multimodal Language Model for Quadruped Robot Learning. *arXiv* **2024**, arXiv:2412.15576.
150. Zheng, R.; Liang, Y.; Huang, S.; Gao, J.; Daumé, H.; Kolobov, A.; Huang, F.; Yang, J. TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies. *arXiv* **2024**, arXiv:2412.10345.
151. Mao, J.; Zhao, S.; Song, S.; Shi, T.; Ye, J.; Zhang, M.; Geng, H.; Malik, J.; Guizilini, V.; Wang, Y. Learning from Massive Human Videos for Universal Humanoid Pose Control. *arXiv* **2024**, arXiv:2412.14172.
152. Zhang, H.; Yu, H.; Zhao, L.; Choi, A.; Bai, Q.; Yang, B.; Xu, W. SLIM: Sim-to-Real Legged Instructive Manipulation via Long-Horizon Visuomotor Learning. *arXiv* **2025**, arXiv:2501.09905.
153. Pertsch, K.; Stachowicz, K.; Ichter, B.; Driess, D.; Nair, S.; Vuong, Q.; Mees, O.; Finn, C.; Levine, S. FAST: Efficient Action Tokenization for Vision-Language-Action Models. *arXiv* **2025**, arXiv:2501.09747.
154. Zhong, Y.; Huang, X.; Li, R.; Zhang, C.; Chen, Z.; Guan, T.; Zeng, F.; Lui, K.N.; Ye, Y.; Liang, Y.; et al. DexGraspVLA: A Vision-Language-Action Framework Towards General Dexterous Grasping. *arXiv* **2025**, arXiv:2502.20900.
155. Wen, J.; Zhu, Y.; Li, J.; Tang, Z.; Shen, C.; Feng, F. DexVLA: Vision-Language Model with Plug-In Diffusion Expert for General Robot Control. *arXiv* **2025**, arXiv:2502.05855.
156. Shi, L.X.; Ichter, B.; Equi, M.; Ke, L.; Pertsch, K.; Vuong, Q.; Tanner, J.; Walling, A.; Wang, H.; Fusai, N.; et al. Hi Robot: Open-Ended Instruction Following with Hierarchical Vision-Language-Action Models. *arXiv* **2025**, arXiv:2502.19417.
157. Ding, P.; Ma, J.; Tong, X.; Zou, B.; Luo, X.; Fan, Y.; Wang, T.; Lu, H.; Mo, P.; Liu, J.; et al. Humanoid-VLA: Towards Universal Humanoid Control with Visual Integration. *arXiv* **2025**, arXiv:2502.14795.
158. Yang, J.; Tan, R.; Wu, Q.; Zheng, R.; Peng, B.; Liang, Y.; Gu, Y.; Cai, M.; Ye, S.; Jang, J.; et al. Magma: A Foundation Model for Multimodal AI Agents. *arXiv* **2025**, arXiv:2502.13130.

159. Jiang, Y.; Zhang, R.; Wong, J.; Wang, C.; Ze, Y.; Yin, H.; Gokmen, C.; Song, S.; Wu, J.; Fei-Fei, L. BEHAVIOR Robot Suite: Streamlining Real-World Whole-Body Manipulation for Everyday Household Activities. *arXiv* **2025**, arXiv:2503.05652.
160. Zhao, Q.; Lu, Y.; Kim, M.J.; Fu, Z.; Zhang, Z.; Wu, Y.; Li, Z.; Ma, Q.; Han, S.; Finn, C.; et al. CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models. *arXiv* **2025**, arXiv:2503.22020.
161. Su, Y.; Zhan, X.; Fang, H.; Xue, H.; Fang, H.S.; Li, Y.L.; Lu, C.; Yang, L. Dense Policy: Bidirectional Autoregressive Learning of Actions. *arXiv* **2025**, arXiv:2503.13217.
162. Figure AI. Natural Humanoid Walk Using Reinforcement Learning. Available online: https://www.figure.ai/news/reinforcement-learning-walking (accessed on 6 October 2026).
163. Qiu, R.Z.; Yang, S.; Cheng, X.; Chawla, C.; Li, J.; He, T.; Yan, G.; Yoon, D.J.; Hoque, R.; Paulsen, L.; et al. Humanoid Policy ∼ Human Policy. *arXiv* **2025**, arXiv:2503.13441.
164. Liu, J.; Chen, H.; An, P.; Liu, Z.; Zhang, R.; Gu, C.; Li, X.; Guo, Z.; Chen, S.; Liu, M.; et al. HybridVLA: Collaborative Diffusion and Autoregression in a Unified Vision-Language-Action Model. *arXiv* **2025**, arXiv:2503.10631.
165. Hung, C.Y.; Sun, Q.; Hong, P.; Zadeh, A.; Li, C.; Tan, U.X.; Majumder, N.; Poria, S. NORA: A Small Open-Sourced Generalist Vision Language Action Model for Embodied Tasks. *arXiv* **2025**, arXiv:2504.19854.
166. Li, J.; Cheng, X.; Huang, T.; Yang, S.; Qiu, R.Z.; Wang, X. AMO: Adaptive Motion Optimization for Hyper-Dexterous Humanoid Whole-Body Control. *arXiv* **2025**, arXiv:2505.03738.
167. NVIDIA. GR00T N1.5. Available online: https://research.nvidia.com/labs/gear/gr00t-n1_5/ (accessed on 6 October 2026).
168. Deng, S.; Yan, M.; Wei, S.; Ma, H.; Yang, Y.; Chen, J.; Zhang, Z.; Yang, T.; Zhang, X.; Zhang, W.; et al. GraspVLA: a Grasping Foundation Model Pre-trained on Billion-scale Synthetic Action Data. *arXiv* **2025**, arXiv:2505.03233.
169. Cai, W.; Peng, J.; Yang, Y.; Zhang, Y.; Wei, M.; Wang, H.; Chen, Y.; Wang, T.; Pang, J. NavDP: Learning Sim-to-Real Navigation Diffusion Policy with Privileged Information Guidance. *arXiv* **2025**, arXiv:2505.08712.
170. Ze, Y.; Chen, Z.; Araújo, J.P.; ang Cao, Z.; Peng, X.B.; Wu, J.; Liu, C.K. TWIST: Teleoperated Whole-Body Imitation System. *arXiv* **2025**, arXiv:2505.02833.
171. Bu, Q.; Yang, Y.; Cai, J.; Gao, S.; Ren, G.; Yao, M.; Luo, P.; Li, H. UniVLA: Learning to Act Anywhere with Task-centric Latent Actions. *arXiv* **2025**, arXiv:2505.06111.
172. Driess, D.; Springenberg, J.T.; Ichter, B.; Yu, L.; Li-Bell, A.; Pertsch, K.; Ren, A.Z.; Walke, H.; Vuong, Q.; Shi, L.X.; et al. Knowledge Insulating Vision-Language-Action Models: Train Fast, Run Fast, Generalize Better. *arXiv* **2025**, arXiv:2505.23705.
173. 1X Technologies. Redwood AI. Available online: https://www.1x.tech/discover/redwood-ai (accessed on 6 October 2026).
174. Wang, Y.; Yang, M.; Ding, Z.; Zhang, Y.; Zeng, W.; Xu, X.; Jiang, H.; Lu, Z. From Experts to a Generalist: Toward General Whole-Body Control for Humanoid Robots. *arXiv* **2025**, arXiv:2506.12779.
175. Zhang, W.; Hu, T.; Zhang, H.; Qiao, Y.; Qin, Y.; Li, Y.; Liu, J.; Kong, T.; Liu, L.; Ma, X. Chain-of-Action: Trajectory Autoregressive Modeling for Robotic Manipulation. *arXiv* **2025**, arXiv:2506.09990.
176. Dyna Robotics. Dynamism v1 (Dyna-1) Model: A Breakthrough in Performance and Production-Ready Embodied AI. Available online: https://www.dyna.co/research/dyna-1 (accessed on 6 October 2026).
177. Google DeepMind. Gemini Robotics On-Device brings AI to local robotic devices. Available online: https://deepmind.google/discover/blog/gemini-robotics-on-device-brings-ai-to-local-robotic-devices/ (accessed on 6 October 2026).
178. Chen, Z.; Ji, M.; Cheng, X.; Peng, X.; Peng, X.B.; Wang, X. GMT: General Motion Tracking for Humanoid Whole-Body Control. *arXiv* **2025**, arXiv:2506.14770.
179. Shukor, M.; Aubakirova, D.; Capuano, F.; Kooijmans, P.; Palma, S.; Zouitine, A.; Aractingi, M.; Pascal, C.; Russi, M.; Marafioti, A.; et al. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics. *arXiv* **2025**, arXiv:2506.01844.
180. Wang, Y.; Li, X.; Wang, W.; Zhang, J.; Li, Y.; Chen, Y.; Wang, X.; Zhang, Z. Unified Vision-Language-Action Model. *arXiv* **2025**, arXiv:2506.19850.
181. Cen, J.; Yu, C.; Yuan, H.; Jiang, Y.; Huang, S.; Guo, J.; Li, X.; Song, Y.; Luo, H.; Wang, F.; et al. WorldVLA: Towards Autoregressive Action World Model. *arXiv* **2025**, arXiv:2506.21539.
182. Chen, S.; Liu, J.; Qian, S.; Jiang, H.; Li, L.; Zhang, R.; Liu, Z.; Gu, C.; Hou, C.; Wang, P.; et al. AC-DiT: Adaptive Coordination Diffusion Transformer for Mobile Manipulation. *arXiv* **2025**, arXiv:2507.01961.
183. Luo, H.; Feng, Y.; Zhang, W.; Zheng, S.; Wang, Y.; Yuan, H.; Liu, J.; Xu, C.; Jin, Q.; Lu, Z. Being-H0: Vision-Language-Action Pretraining from Large-Scale Human Videos. *arXiv* **2025**, arXiv:2507.15597.
184. Yang, R.; Yu, Q.; Wu, Y.; Yan, R.; Li, B.; Cheng, A.C.; Zou, X.; Fang, Y.; Cheng, X.; Qiu, R.Z.; et al. EgoVLA: Learning Vision-Language-Action Models from Egocentric Human Videos. *arXiv* **2025**, arXiv:2507.12440.
185. Cheang, C.; Chen, S.; Cui, Z.; Hu, Y.; Huang, L.; Kong, T.; Li, H.; Li, Y.; Liu, Y.; Ma, X.; et al. GR-3 Technical Report. *arXiv* **2025**, arXiv:2507.15493.
186. Shanghai AI Laboratory. InternVLA-N1: An Open Dual-System Vision-Language Navigation Foundation Model with Learned Latent Plans. Available online: https://internrobotics.github.io/internvla-n1.github.io/static/pdfs/InternVLA_N1.pdf (accessed on 14 October 2026).

187. Skild AI. Building the general-purpose robotic brain. Available online: https://www.skild.ai/blogs/building-the-general-purpose-robotic-brain (accessed on 6 October 2026).
188. Wei, M.; Wan, C.; Yu, X.; Wang, T.; Yang, Y.; Mao, X.; Zhu, C.; Cai, W.; Wang, H.; Chen, Y.; et al. StreamVLN: Streaming Vision-and-Language Navigation via SlowFast Context Modeling. *arXiv* **2025**, arXiv:2507.05240.
189. TRI LBM Team.; Barreiros, J.; Beaulieu, A.; Bhat, A.; Cory, R.; Cousineau, E.; Dai, H.; Fang, C.H.; Hashimoto, K.; Irshad, M.Z.; et al. A Careful Examination of Large Behavior Models for Multitask Dexterous Manipulation. *arXiv* **2025**, arXiv:2507.05331.
190. Chen, X.; Wei, H.; Zhang, P.; Zhang, C.; Wang, K.; Guo, Y.; Yang, R.; Wang, Y.; Xiao, X.; Zhao, L.; et al. villa-X: Enhancing Latent Action Modeling in Vision-Language-Action Models. *arXiv* **2025**, arXiv:2507.23682.
191. Boston Dynamics; Toyota Research Institute. Large Behavior Models and Atlas Find New Footing. Available online: https://bostondynamics.com/blog/large-behavior-models-atlas-find-new-footing/ (accessed on 6 October 2026).
192. Qu, D.; Song, H.; Chen, Q.; Chen, Z.; Gao, X.; Wang, D.; Ye, X.; Lv, Q.; Shi, M.; Ren, G.; et al. EO-1: An Open Unified Embodied Foundation Model for General Robot Control. *arXiv* **2025**, arXiv:2508.21112.
193. Jiang, T.; Yuan, T.; Liu, Y.; Lu, C.; Cui, J.; Liu, X.; Cheng, S.; Gao, J.; Xu, H.; Zhao, H. Galaxea Open-World Dataset and G0 Dual-System VLA Model. *arXiv* **2025**, arXiv:2509.00576.
194. Liao, Y.; Zhou, P.; Huang, S.; Yang, D.; Chen, S.; Jiang, Y.; Hu, Y.; Cai, J.; Liu, S.; Luo, J.; et al. Genie Envisioner: A Unified World Foundation Platform for Robotic Manipulation. *arXiv* **2025**, arXiv:2508.05635.
195. Lee, J.; Duan, J.; Fang, H.; Deng, Y.; Liu, S.; Li, B.; Fang, B.; Zhang, J.; Wang, Y.R.; Lee, S.; et al. MolmoAct: Action Reasoning Models that can Reason in Space. *arXiv* **2025**, arXiv:2508.07917.
196. Jiang, Y.; Huang, S.; Xue, S.; Zhao, Y.; Cen, J.; Leng, S.; Li, K.; Guo, J.; Wang, K.; Chen, M.; et al. RynnVLA-001: Using Human Demonstrations to Improve Robot Manipulation. *arXiv* **2025**, arXiv:2509.15212.
197. Lv, Q.; Kong, W.; Li, H.; Zeng, J.; Qiu, Z.; Qu, D.; Song, H.; Chen, Q.; Deng, X.; Pang, J. F1: A Vision-Language-Action Model Bridging Understanding and Generation to Actions. *arXiv* **2025**, arXiv:2509.06951.
198. Reuss, M.; Zhou, H.; Rühle, M.; Ömer Erdinç Yağmurlu.; Otto, F.; Lioutikov, R. FLOWER: Democratizing Generalist Robot Policies with Efficient Vision-Language-Action Flow Policies. *arXiv* **2025**, arXiv:2509.04996.
199. Gemini Robotics Team.; Abdolmaleki, A.; Abeyruwan, S.; Ainslie, J.; Alayrac, J.B.; Arenas, M.G.; Balakrishna, A.; Batchelor, N.; Bewley, A.; Bingham, J.; et al. Gemini Robotics 1.5: Pushing the Frontier of Generalist Robots with Advanced Embodied Reasoning, Thinking, and Motion Transfer. *arXiv* **2025**, arXiv:2510.03342.
200. NVIDIA. GR00T N1.6. Available online: https://research.nvidia.com/labs/gear/gr00t-n1_6/ (accessed on 6 October 2026).
201. Zhang, J.; Li, A.; Qi, Y.; Li, M.; Liu, J.; Wang, S.; Liu, H.; Zhou, G.; Wu, Y.; Li, X.; et al. Embodied Navigation Foundation Model. *arXiv* **2025**, arXiv:2509.12129.
202. Hirose, N.; Glossop, C.; Shah, D.; Levine, S. OmniVLA: An Omni-Modal Vision-Language-Action Model for Robot Navigation. *arXiv* **2025**, arXiv:2509.19480.
203. Liu, S.; Li, B.; Ma, K.; Wu, L.; Tan, H.; Ouyang, X.; Su, H.; Zhu, J. RDT2: Exploring the Scaling Limit of UMI Data Towards Zero-Shot Cross-Embodiment Generalization. *arXiv* **2026**, arXiv:2602.03310.
204. Unitree. UnifoLM-WMA-0 (GitHub repository unitreerobotics/unifolm-world-model-action). Available online: https://github.com/unitreerobotics/unifolm-world-model-action (accessed on 6 October 2026).
205. Wang, Y.; Ding, P.; Li, L.; Cui, C.; Ge, Z.; Tong, X.; Song, W.; Zhao, H.; Zhao, W.; Hou, P.; et al. VLA-Adapter: An Effective Paradigm for Tiny-Scale Vision-Language-Action Model. *arXiv* **2025**, arXiv:2509.09372.
206. Zhai, A.; Liu, B.; Fang, B.; Cai, C.; Ma, E.; Yin, E.; Wang, H.; Zhou, H.; Wang, J.; Shi, L.; et al. Igniting VLMs toward the Embodied Space. *arXiv* **2025**, arXiv:2509.11766.
207. GigaBrain Team.; Ye, A.; Wang, B.; Ni, C.; Huang, G.; Zhao, G.; Li, H.; Li, J.; Zhu, J.; Feng, L.; et al. GigaBrain-0: A World Model-Powered Vision-Language-Action Model. *arXiv* **2025**, arXiv:2510.19430.
208. Chen, X.; Chen, Y.; Fu, Y.; Gao, N.; Jia, J.; Jin, W.; Li, H.; Mu, Y.; Pang, J.; Qiao, Y.; et al. InternVLA-M1: A Spatially Guided Vision-Language-Action Framework for Generalist Robot Policy. *arXiv* **2025**, arXiv:2510.13778.
209. Goyal, A.; Hadfield, H.; Yang, X.; Blukis, V.; Ramos, F. VLA-0: Building State-of-the-Art VLAs with Zero Modification. *arXiv* **2025**, arXiv:2510.13054.
210. Zheng, J.; Li, J.; Wang, Z.; Liu, D.; Kang, X.; Feng, Y.; Zheng, Y.; Zou, J.; Chen, Y.; Zeng, J.; et al. X-VLA: Soft-Prompted Transformer as Scalable Cross-Embodiment Vision-Language-Action Model. *arXiv* **2025**, arXiv:2510.10274.
211. Sunday Robotics. ACT-1: A Robot Foundation Model Trained on Zero Robot Data. Available online: https://www.sunday.ai/blog/no-robot-data (accessed on 14 October 2026).
212. Lin, T.; Zhong, Y.; Du, Y.; Zhang, J.; Liu, J.; Chen, Y.; Gu, E.; Liu, Z.; Cai, H.; Zou, Y.; et al. Evo-1: Lightweight Vision-Language-Action Model with Preserved Semantic Alignment. *arXiv* **2025**, arXiv:2511.04555.
213. Hung, C.Y.; Majumder, N.; Deng, H.; Renhang, L.; Ang, Y.; Zadeh, A.; Li, C.; Herremans, D.; Wang, Z.; Poria, S. NORA-1.5: A Vision-Language-Action Model Trained using World Model- and Action-based Preference Rewards. *arXiv* **2025**, arXiv:2511.14659.

214. Ze, Y.; Zhao, S.; Wang, W.; Kanazawa, A.; Duan, R.; Abbeel, P.; Shi, G.; Wu, J.; Liu, C.K. TWIST2: Scalable, Portable, and Holistic Humanoid Data Collection System. *arXiv* **2025**, arXiv:2511.02832.
215. He, T.; Wang, Z.; Xue, H.; Ben, Q.; Luo, Z.; Xiao, W.; Yuan, Y.; Da, X.; Castañeda, F.; Sastry, S.; et al. VIRAL: Visual Sim-to-Real at Scale for Humanoid Loco-Manipulation. *arXiv* **2025**, arXiv:2511.15200.
216. Physical Intelligence. $\pi_{0.6}$ Model Card. Available online: https://website.pi-asset.com/pi06star/PI06_model_card.pdf (accessed on 6 October 2026).
217. Wen, R.; Chen, G.; Cui, Z.; Du, M.; Gou, Y.; Han, Z.; Huang, L.; Lei, M.; Li, Y.; Li, Z.; et al. GR-Dexter Technical Report. *arXiv* **2025**, arXiv:2512.24210.
218. Jiang, H.; Chen, J.; Bu, Q.; Chen, L.; Shi, M.; Zhang, Y.; Li, D.; Suo, C.; Wang, C.; Peng, Z.; et al. WholeBodyVLA: Towards Unified Latent VLA for Whole-Body Loco-Manipulation Control. *arXiv* **2025**, arXiv:2512.11047.
219. Black, K.; Ren, A.Z.; Equi, M.; Levine, S. Training-Time Action Conditioning for Efficient Real-Time Chunking. *arXiv* **2025**, arXiv:2512.05964.
220. 1X Technologies. 1X World Model: From Video to Action: A New Way Robots Learn. Available online: https://www.1x.tech/discover/world-model-self-learning (accessed on 6 October 2026).
221. Luo, H.; Wang, Y.; Zhang, W.; Zheng, S.; Xi, Z.; Xu, C.; Xu, H.; Yuan, H.; Zhang, C.; Wang, Y.; et al. Being-H0.5: Scaling Human-Centric Robot Learning for Cross-Embodiment Generalization. *arXiv* **2026**, arXiv:2601.12993.
222. Galaxea AI. GalaxeaVLA: G0Plus (GitHub repository OpenGalaxea/GalaxeaVLA). Available online: https://github.com/OpenGalaxea/GalaxeaVLA (accessed on 6 October 2026).
223. Figure AI. Introducing Helix 02: Full-Body Autonomy. Available online: https://www.figure.ai/news/helix-02 (accessed on 6 October 2026).
224. Cai, J.; Cai, Z.; Cao, J.; Chen, Y.; He, Z.; Jiang, L.; Li, H.; Li, H.; Li, Y.; Liu, Y.; et al. InternVLA-A1: Unifying Understanding, Generation and Action for Robotic Manipulation. *arXiv* **2026**, arXiv:2601.02456.
225. Wu, W.; Lu, F.; Wang, Y.; Yang, S.; Liu, S.; Wang, F.; Zhu, Q.; Sun, H.; Wang, Y.; Ma, S.; et al. A Pragmatic VLA Foundation Model. *arXiv* **2026**, arXiv:2601.18692.
226. Microsoft Research. Introducing Rho-alpha, the new robotics model from Microsoft. Available online: https://www.microsoft.com/en-us/research/?p=1159224 (accessed on 6 October 2026).
227. Spirit AI. Spirit-v1.5 (Hugging Face model card). Available online: https://huggingface.co/Spirit-AI-robotics/Spirit-v1.5 (accessed on 6 October 2026).
228. Unitree. UnifoLM-VLA-0 (GitHub repository unitreerobotics/unifolm-vla). Available online: https://raw.githubusercontent.com/unitreerobotics/unifolm-vla/main/README.md (accessed on 6 October 2026).
229. Zheng, R.; Niu, D.; Xie, Y.; Wang, J.; Xu, M.; Jiang, Y.; Castañeda, F.; Hu, F.; Tan, Y.L.; Fu, L.; et al. EgoScale: Scaling Dexterous Manipulation with Diverse Egocentric Human Data. *arXiv* **2026**, arXiv:2602.16710.
230. Galaxea AI. GalaxeaVLA: G0Tiny (GitHub repository OpenGalaxea/GalaxeaVLA). Available online: https://github.com/OpenGalaxea/GalaxeaVLA (accessed on 6 October 2026).
231. Lin, X.; Lin, T.; Du, Y.; Xie, H.; Jin, Y.; Li, J.; Wu, S.; Wang, Q.; Li, M.; Zhao, M.; et al. HoloBrain-0 Technical Report. *arXiv* **2026**, arXiv:2602.12062.
232. Cai, R.; Guo, J.; He, X.; Jin, P.; Li, J.; Lin, B.; Liu, F.; Liu, W.; Ma, F.; Ma, K.; et al. Xiaomi-Robotics-0: An Open-Sourced Vision-Language-Action Model with Real-Time Execution. *arXiv* **2026**, arXiv:2602.12684.
233. Tu, R.; Shukla, A.; Yoo, S.; Li, X.; Li, J.; Xie, J.; Su, H.; Tu, Z. SG-VLA: Learning Spatially-Grounded Vision-Language-Action Models for Mobile Manipulation. *arXiv* **2026**, arXiv:2603.22760.
234. Wei, S.; Jing, H.; Li, B.; Zhao, Z.; Mao, J.; Ni, Z.; He, S.; Liu, J.; Liu, X.; Kang, K.; et al. Ψ_0: An Open Foundation Model Towards Universal Humanoid Loco-Manipulation. *arXiv* **2026**, arXiv:2603.12263.
235. Luo, H.; Zhang, W.; Feng, Y.; Zheng, S.; Xu, H.; Xu, C.; Xi, Z.; Fu, Y.; Lu, Z. Being-H0.7: A Latent World-Action Model from Egocentric Videos. *arXiv* **2026**, arXiv:2605.00078.
236. AgiBot. The Unity of Reasoning and Action: AGIBOT Unveils Genie Operator. Available online: https://agibot.com/article/231/detail/56.html (accessed on 6 October 2026).
237. Fang, H.; Duan, J.; Clay, D.; Wang, S.; Liu, S.; Huang, W.; Fan, X.; Tsai, W.C.; Chen, S.; Wang, Y.R.; et al. MolmoAct2: Action Reasoning Models for Real-world Deployment. *arXiv* **2026**, arXiv:2605.02881.
238. Ding, H.; Zhang, S.; Xu, Z.; Guo, J.; Liu, H.; Cheng, X.; Chen, Z.; Qi, H.; Wang, D.; Xu, H.; et al. Uni-LaViRA: Language-Vision-Robot Actions Translation for Unified Embodied Navigation. *arXiv* **2026**, arXiv:2605.27582.
239. Zhang, L.; Gong, Z.; Hao, X.; Fu, H.; Zhang, Q.; Zhou, M.; Ye, H.; Liang, X.; Liang, J.; Ding, W. FutureNav: Unified World-Action Modeling for Vision-and-Language Navigation. *arXiv* **2026**, arXiv:2606.30367.
240. Liu, Y.; Dong, Z.; Ye, B.; Yuan, T.; Jiang, T.; Yang, A.; Cao, S.; Liu, H.; Sun, Y.; Guo, Z.; et al. G0.5: One Autoregressive Stream for Robot Reasoning and Action. *arXiv* **2026**, arXiv:2608.11739.

241. Xinhua News Agency. Galbot AstraBrain report (in Chinese). Available online: https://www.xinhuanet.com/digital/20260821/e172222483dc4d75a357145caa74d9d0/c.html (accessed on 6 October 2026).
242. Hu, Y.; Zhu, H.; Zheng, B.; Hu, Y.; Zhang, T.; Chen, Z.; Zhao, J.; Nai, R.; Gao, Y. OpenHLM: An Empirical Recipe for Whole-Body Humanoid Loco-Manipulation. *arXiv* **2026**, arXiv:2606.22174.
243. Zhang, J.; Zhou, G.; Yin, H.; Huang, Y.; Lei, Z.; Peng, Q.; Yuan, H.; Zhang, J.; Guo, X.; Chen, X.; et al. Qwen-RobotNav Technical Report: A Scalable Navigation Model Designed for an Agentic Navigation System. *arXiv* **2026**, arXiv:2606.18112.
244. Google DeepMind. Gemini Robotics 2: Safety Evaluations. Available online: https://storage.googleapis.com/deepmind-media/gemini-robotics/Gemini-Robotics-2-Safety.pdf (accessed on 6 October 2026).
245. Google DeepMind. Gemini Robotics On-Device 2 - Model Card. Available online: https://deepmind.google/models/model-cards/gemini-robotics-on-device-2/ (accessed on 6 October 2026).
246. Ma, H.; Cai, J.; Xu, X.; Li, H.; Yang, Y.; Tian, Y.; Cao, J.; Zhu, H.; Qiu, Z.; Zhaxizhuoma.; et al. InternVLA-A1.5: Unifying Understanding, Latent Foresight, and Action for Compositional Generalization. *arXiv* **2026**, arXiv:2607.04988.
247. Wu, W.; Wang, F.; Lu, F.; Sun, H.; Liu, S.; Wang, Y.; Yan, Y.; Wang, Y.; Ma, S.; Wang, X.; et al. From Foundation to Application: Improving VLA Models in Practice. *arXiv* **2026**, arXiv:2607.06403.
248. Xiaomi Robotics Team.; Guo, J.; Jin, P.; Li, J.; Li, P.; Li, Y.; Liu, F.; Peng, W.; Qin, O.; Su, Y.; et al. Xiaomi-Robotics-1: Scaling Vision-Language-Action Models with over 100K Hours of Real-World Trajectories. *arXiv* **2026**, arXiv:2607.15330.
249. Yang, D.; Chen, H.; Chen, X.; Liu, L.; Li, M.; Tu, C.; Xu, K.; Ma, X.; Liu, S. Learning Panorama-Aware VLA for Mobile Manipulation with Whole-Body Teleoperation. *arXiv* **2026**, arXiv:2608.02257.
250. Skild AI. Introducing S1: In-Context Learning for Robotics. Available online: https://www.skild.ai/blogs/s1 (accessed on 6 October 2026).
251. Cai, X.; Cai, Y.; Chen, B.; Chen, J.; Chen, Z.; Feng, S.; Hou, T.; Huang, J.; Jiang, H.; Ju, R.; et al. τ_0-VLA: a Hierarchical Robot Foundation Model with World-Model-Guided Test-Time Computation. *arXiv* **2026**, arXiv:2608.16885.
252. Li, Z.; Zhang, Z.; Wei, Y.; Zhang, W.; Yuan, X.; Zhi, P.; Li, G.; Guo, X.; Gao, F.; Yang, J.; et al. ω-0: A Latent Predictive World Action Model for Concurrent Humanoid Loco-Manipulation. *arXiv* **2026**, arXiv:2608.06375.
253. Chen, T.; Lin, W.; Lyu, C.; Xiang, P.; Tang, J.; Nian, T.; Cai, Z.; Ye, K.; Yu, C.; Zhu, W.; et al. Spark-0: Scaling Spatially Aligned Tactile Pretraining for Robot Dexterous Manipulation. *arXiv* **2026**, arXiv:2610.22342.